

CONTINUATION.md — vasic-digital tmux

Last updated: 2026-09-01T16:15:00Z

§0 — How to resume work in any CLI agent

Paste this prompt:

Read CONTINUATION.md at the repo root. Identify the topmost item under §3 Active work with status IN PROGRESS or BLOCKED. Re-read Constitution.md, CLAUDE.md, AGENTS.md, Issues.md, Fixed.md for mandates and current backlog. Resume from current state. Update this document as you work.

§1 — Snapshot

Field	Value
Repo	vasic-digital/tmux on GitHub + GitLab
Origin	Migrated from ATMOSphere project (scripts/tmux/, docker/Dockerfile.tmux-build, docs/guides/TMUX_OPTIMIZED_BUILD.md) on 2026-05-07
Pinned tmux	upstream release TAG 3.7b (e802909de06012a4df6209d55e86487c56223163) — re-pinned by TMX-089 (a6f3fc4) from the untagged next-3.8 (40381bdc) that 562cde7 had briefly adopted. 3.7b is an ANCESTOR of 40381bdc (verified — a clean rewind, not a divergence); its release tarball exists with measured sha256 87f2e99e3b685973f2ca002ffd6ed7e51a5744f7009daae5a15670b6d532db96 and ships a pre-generated configure AND cmd-parse.c, which is what the root-free zig build path requires. Rebuilt binary reports tmux 3.7b (verified). Note: tag 3.7c EXISTS upstream, so 3.7b is NOT the latest stable and README.md no longer claims it is.
Version	1.0.44 (versionCode 45, NOT yet released — no tag cut) — FOUR items this cycle: TMX-086 + TMX-087 (the operator-reported install

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	blocker) and, landing after that batch already reached main, TMX-088 (the zig build path's YACC was an ABSOLUTE path, breaking BOTH configure's AC_CHECK_PROG resolution AND etc/ylwrap's output-filename contract — the latter a SILENT exit-0 failure; found by INDEPENDENT post-hoc review of 562cde7, fixed in 7966edb, new test 91 + mutation M-ZIG-YACC-BARE-WORD) and TMX-089 (562cde7's next-3.8 pin withdrew the root-free build capability — test 71 C4 MEASURED the loss; re-pinned to release TAG 3.7b in a6f3fc4 on a second explicit operator decision the same day, tarball fast-path restored but now DERIVED from the submodule's configure.ac and refused without a matching pinned sha256). See §3.36 (cycle state) + Fixed.md §M (TMX-088/089). TMX-086 + TMX-087: TWO INDEPENDENT root causes behind the operator's single "cannot install / setup tmx" report, each one alone sufficient to block the install — (1) install.sh injected an HTTPS instead of rewrite BY DEFAULT, which git propagates into every recursive submodule clone via GIT_CONFIG_PARAMETERS, so it reached the one PRIVATE nested repo (HelixDevelopment/helix_perf_cache) and blocked forever on a GitHub Username prompt with no GIT_TERMINAL_PROMPT=0 guard; (2) the constitution submodule's index carried a .gitmodules-unmapped duplicate design-toolkit gitlink, aborting every consumer's recursive clone with Failed to recurse into submodule path 'constitution'. Fixed by defaulting the clone URL to SSH + making the rewrite an explicit opt-in (TMX_INSTALL_HTTPS_REWRITE=1) + exporting the no-prompt / no-hang git env (GIT_TERMINAL_PROMPT=0, GIT_ASKPASS="", SSH_ASKPASS="", GIT_SSH_COMMAND with accept-new), and by constitution commit b9096ac (orphan gitlink removed, ff-pushed to all six upstreams, no force-push); constitution pointer bumped 6f24df9 → b9096ac. New test 90 (§11.4.115 RED/GREEN) + 3 paired mutations. See §3.35 + Fixed.md §L (TMX-086/087); §3.36 + Fixed.md §M (TMX-088/089). Previous: v1.0.43 (TMX-084 + TMX-085 — two false-result defects in the verification HARNESS itself, not the wrapper: run_all.sh's verdict classifier misread an honest SKIP-layer: sub-check note (and any test's own internal PASS=N FAIL=N SKIP=N counters) as the test's overall verdict, and test 59's RED_MODE default made it fail 100% deterministically against a genuinely-fixed wrapper under a bare harness invocation — Fixed.md §K); v1.0.42 (TMX-083, systemd-oomd victim-avoidance on session scopes — ManagedOOMPreference=avoid added to every scope/slice creation site, closing a §11.4.108 layer-3 gap where MemoryMax=infinity was correct but ineffective against oomd's orthogonal PSI-pressure victim selection; renumbered from v1.0.41 to v1.0.42 per §11.4.113 merge-onto-latest-main after a

Field	Value
Verification (this cycle, v1.0.44)	concurrent main branch shipped a separate v1.0.41 install-verification-RED batch — see Fixed.md §J, TMX-083); v1.0.41 (install-verification-RED fix — 4 distinct root causes across tests 22/58/77/86/88, including a v1.0.40 documentation-layer bluff where the opcode.json fix was CLAIMED but never actually landed — self-caught and corrected); v1.0.40 (TMX-082, split-topology test reconciliation); v1.0.39 (TMX-079, no resource or lifetime limit by default — root cause of "sessions/processes killed despite ample host capacity"); v1.0.38 (TMX-076/077, scope-independent collision guard + meta-test mutations); v1.0.37 (host-adaptive CPUQuota default, interactive server-scope split TMX_SERVER_SPLIT=1, TMX_CLASSIFICATION export, OOM score adjustment). Controlled A/B for TMX-086 (same repo, same host, same minute, ONE variable): SSH as declared → exit=0, 6adcfb2 refs/heads/main; with the installer's insteadOf rewrite → exit=128, fatal: could not read Username for 'https://github.com'. Anonymous-HTTPS census (all git config bypassed, no credential helper on host): 9 repos PUBLIC, HelixDevelopment/helix_perf_cache the sole PRIVATE one — the trigger. New test 90 (90_install_ssh_only_no_credential_prompt.sh) drives the real installer behind a recording git shim that captures the REAL argv + the INHERITED GIT_TERMINAL_PROMPT, so its assertions are runtime evidence, not a grep of the source (§11.4.226): GREEN RED_MODE=0 → PASS=4 FAIL=0 SKIP=0 exit 0; RED RED_MODE=1 (reconstructed pre-fix installer) → PASS=1 FAIL=0 SKIP=0 exit 0, defect reproduced. TMX-087: with the orphan gitlink removed, a recursive clone checked out Containers and constitution over SSH with NO credential prompt; constitution b9096ac ff-pushed (6f24df9..b9096ac) to all six upstreams, no force-push. A self-inflicted test-instrument FAIL-bluff (argv extractor keyed on ' clone' with a leading space, matching only when the defect was present) was found and fixed at the instrument layer during authoring per §11.4.1. TMX-088 measured: AC_CHECK_PROG with ac_word=/usr/bin/bison → found_yacc=no, with ac_word=bison → yes, bogus-name negative control → no; real tmux/etc/ylwrap with /usr/bin/bison → rc=0 and cmd-parse.c MISSING, with bison -y → rc=0 and cmd-parse.c CREATED (both exit 0 — an exit-status assertion could never have caught it). New test 91 (91_zig_yacc_contract.sh, §11.4.115 RED/GREEN, RED_MODE default 0) exercises BOTH real contracts without a full build and DERIVES the value under test from build_native.sh itself (§11.4.201(6) attributability); paired mutation M-ZIG-YACC-BARE-WORD verified to make C1+C2 FAIL. TMX-089 measured: under the

Field	Value
	next-3.8 pin test 71 C4 FAILED on its neutered bare host (cmd-parse.c is not pre-generated and no yacc/bison found) — the measured capability withdrawal; 3.7b verified an ancestor of 40381bdc; tarball sha256 87f2e99e...d532db96 measured + pinned; rebuilt binary reports tmux 3.7b; tag 3.7c confirmed to exist. KNOWN-OPEN — explicitly NOT claimed (§11.4.6): (1) full-suite run_all.sh retest UNCONFIRMED: — has NOT completed, outcome not known, still owed before the tag (§11.4.40); (2) the full meta-test mutation sweep has NOT been run this cycle — individual mutations verified, whole-sweep result not known; (3) test 71 C4 has NOT been re-run on the 3.7b artifact — the root-free restoration is the RATIONALE for the re-pin, UNCONFIRMED: as a result; (4) test 68 (68_session_lifecycle.sh) C6/C7 FAIL — pre-existing, reproduces identically on the pre-change tree, TMX-080/TMX-081 family (Issues.md §H1/§H2), investigation in flight, no outcome recorded; (5) FAIL: T3: live WheelUpPane binding is not the copy-mode override (17_scrollback_copy_mode.sh) — real, currently failing, newly tracked as TMX-090 (Issues.md §I1); under investigation, cause NOT established and none asserted; (6) .html/.pdf twins STALE (pandoc/weasyprint absent), remediation attempt in flight — an honest stale, never a faked transform (§11.4.106(E)). Process note (§11.4.142/§11.4.113): 562cde7 swept an agent's in-flight build_native.sh edit onto main before the mandatory independent review completed (commit_all.sh uses git add -A); remediated FIX-FORWARD in 7966edb, no history rewrite, no force-push. Full detail: Fixed.md §L + §M. (Prior cycle, v1.0.43: investigated all 5 operator-reported failures (27_state_persistence.sh 56_real_mouse_drag_copy.sh 57_reload_select_copy_paste.sh 59_oomd_preference_avoid.sh 87_server_scope_split.sh) via /superpowers:systematic-debugging; 27/87 confirmed unchanged instances of the already-tracked TMX-080/TMX-081 (no action, still Queued); 56/57/59 root-caused to two harness-side classification bugs (TMX-084/085) — NEITHER was a defect in scripts/tmx/scripts/tmx.template. New regression test 89_classify_verdict_carrier.sh (§11.4.115 RED/GREEN, 6 fixtures) + paired mutations M-CLASSIFY-CARRIER/M-RED-DEFAULT-59. Full detail: Fixed.md §K.) (Prior cycle, v1.0.40: install.sh clean PASS=72 FAIL=0 SKIP=14, real live-session split-topology verification via kernel /proc/<pid>/cgroup. Full detail: Fixed.md §I, TMX-082.)
Governance docs	constitution/ submodule (HelixConstitution, pinned b9096ac as of v1.0.44 — bumped from 6f24df9 by the TMX-087 orphan-gitlink fix); Containers/ submodule (pinned 18ed03d); tmux/ submodule pinned

Field	Value
	to release TAG 3.7b (e802909) as of TMX-089 (a6f3fc4), re-pinned from the untagged next-3.8 (40381bdc) — governance files and every version gate swept next-3.8 → 3.7b; project Constitution.md (Project Articles §101–§109 extends the submodule), CLAUDE.md, AGENTS.md, QWEN.md, GEMINI.md (all lockstep), Issues.md, Fixed.md, this document

§2 — Mandates (canonical authority)

- Constitution.md §1 anti-bluff covenant (verbatim user-mandate quote propagated)
- Constitution.md §11.4.1 FAIL-bluffs equally forbidden
- Constitution.md §11.4.2 recorded-evidence requirement
- Constitution.md §11.4.3 per-host-topology test dispatch
- Constitution.md §11.4.4 test-interrupt-on-discovery + 4-layer test coverage
- Constitution.md §11.4.5 audio + video quality analysis (N/A for tmux scope; principle generalized)
- Constitution.md §11.4.6 — No-guessing mandate (verbatim user-mandate quote)
- Constitution.md §9 absolute data safety
- Constitution.md §12.6 60% host memory budget (per-session TMX_MEM is the enforcement)
- Constitution.md §5 / §12.10 continuation-document sacred invariant

§3 — Active work

§3.36 — v1.0.44 release cycle — consolidated state (TMX-086 / TMX-087 / TMX-088 / TMX-089) → 2026-09-01

Status: IN PROGRESS. All FOUR fixes are landed on main; the release is **NOT tagged** and **MUST NOT** be tagged until the known-open list below is cleared. This entry is the single cycle-state record; §3.35 keeps the deep TMX-086/TMX-087 detail.

Scope of this cycle — four items, two waves:

Item	What	Landed in
TMX-086		562cde7

Item	What	Landed in
TMX-087	install.sh injected an HTTPS insteadOf rewrite BY DEFAULT; git propagated it into every recursive submodule clone via GIT_CONFIG_PARAMETERS, reaching the one PRIVATE nested repo (HelixDevelopment/helix_perf_cache) and blocking forever on a GitHub Username prompt with no GIT_TERMINAL_PROMPT=0 guard.	
	The constitution submodule's index carried a .gitmodules-unmapped duplicate design-toolkit gitlink, aborting every consumer's recursive clone. Fixed constitution-side in b9096ac, ff-pushed to all six upstreams, pointer bumped 6f24df9 → b9096ac.	562cde7 (pointer)
TMX-088	The zig (root-free) build path computed YACC from command -v bison — an ABSOLUTE path — breaking BOTH contracts it feeds: configure's AC_CHECK_PROG ("\$as_dir\$ac_word" → /usr/bin//usr/bin/bison, aborts yacc not found on a host that HAS bison) AND etc/ylwrap's output-filename contract (plain bison emits cmd-parse.tab.c, ylwrap still EXITS 0 — a SILENT failure leaving make with no parser source). Fixed to a bare bison -y. Same commit: the autogen refusal message was UNREACHABLE under set -euo pipefail — fixed with true.	7966edb
	562cde7 had adopted the untagged next-3.8 (40381bdc), which has NO release tarball — so the root-free path had to build from the submodule, needing autotools + bison. Test 71	
TMX-089	C4 MEASURED the withdrawal (cmd-parse.c is not pre-generated and no yacc/bison found), a \$11.4.122 capability loss. Re-pinned to release TAG 3.7b (e802909) on a SECOND explicit operator decision the same day.	a6f3fc4

How the second wave was found (§11.4.238 — the intended direction): neither TMX-088 nor TMX-089 was operator-reported. TMX-088 came from an **independent post-hoc code review** of 562cde7; TMX-089 came from a **test measuring a capability withdrawal**.

Process note (§11.4.142 / §11.4.113) — recorded because it is the causal reason TMX-088 shipped broken: 562cde7 swept an agent's IN-FLIGHT scripts/build_native.sh edit onto main before that edit had passed the mandatory independent review — commit_all.sh stages with git add -A, so unrelated in-flight work in the tree was carried along by a commit that was about something else. Remediated **FIX-FORWARD** in 7966edb per §11.4.113: no history rewrite, no force-push at any point.

Current pin + artifact (verified, not inferred):

- Submodule gitlink → e802909de06012a4df6209d55e86487c56223163;
git -C tmux describe --tags → 3.7b.
- 3.7b is an **ancestor** of 40381bdc — a clean rewind, not a divergence.
- Release tarball sha256
87f2e99e3b685973f2ca002ffd6ed7e51a5744f7009daae5a15670b6d532db96 (measured, pinned in scripts/build_native.sh); ships pre-generated configure AND cmd-parse.c.
- **Rebuilt binary reports tmux 3.7b** (read from the binary itself).
- build_native.sh's tarball fast-path is restored but now DERIVES the version from the submodule's own configure.ac and REFUSES the download unless a sha256 is pinned for exactly that version — so a future re-pin to an untagged commit falls back to the submodule build rather than silently shipping a binary that does not match the pin.
- Governance files and every version gate swept next-3.8 → 3.7b.
README.md no longer claims "latest stable" — tag 3.7c exists (measured).

KNOWN-OPEN RIGHT NOW — none of these is claimed fixed, and each blocks the tag or is tracked (§11.4.6 / §11.4.40):

1. **Full-suite run_all.sh retest — UNCONFIRMED:.** Has NOT completed. Outcome not known, not claimed. Owed before any tag (§11.4.40).
2. **Full meta-test mutation sweep — NOT RUN this cycle.** The individual mutations (M-INSTALL-*, M-ZIG-YACC-BARE-WORD) were verified; the whole-sweep result is not known and is not claimed.
3. **Test 71 C4 has NOT been re-run on the 3.7b artifact.** The root-free restoration is the STATED RATIONALE for the re-pin — it is UNCONFIRMED: as a RESULT until C4 is re-run on a neutered host and

captured. This is the missing GREEN half of TMX-089's RED→GREEN pair.

4. **Test 68 (68_session_lifecycle.sh) C6/C7 FAIL** — pre-existing; fails identically on the pre-change tree, so it is NOT caused by this cycle's work. Belongs to the already-tracked TMX-080 / TMX-081 family (Issues.md §H1/§H2). An investigation is IN FLIGHT; **no outcome is recorded here.**
5. **FAIL: T3: live WheelUpPane binding is not the copy-mode override (17_scrollback_copy_mode.sh)** — a real, currently-failing, previously-unrecorded defect, now tracked as **TMX-090** (Issues.md §I1). Under investigation; **the cause is NOT established and none is asserted.** First concrete step: capture the test's own observed: line (the live binding text), which the entry records as
PENDING_FORENSICS:.
6. **.html / .pdf document twins are STALE** — pandoc / weasyprint are absent on this host; a remediation attempt is in flight. Honest stale per §11.4.106(E), never a faked transform.

Immediate next actions (in order):

1. Re-run test 71 **C4** against the 3.7b artifact and capture the output — closes TMX-089's owed GREEN (item 3).
2. Run the full `run_all.sh` sweep + the full meta-test mutation sweep, capture both (items 1–2).
3. Capture the observed: binding text for TMX-090 and open the systematic-debugging arc (item 5).
4. Backfill the `Fixed.md` §L placeholder closure SHAs (see §3.35's "Placeholder owed"). §M already carries real SHAs (7966edb, a6f3fc4).
5. Only after 1–4: cut the tag (§11.4.40 full retest GREEN + §11.4.185 manual QA remain preconditions).

Full detail + captured evidence: `Fixed.md` §L (TMX-086, TMX-087) and §M (TMX-088, TMX-089); `CHANGELOG.md` v1.0.44; `Issues.md` §I1 (TMX-090).

§3.35 — v1.0.44: SSH-only installability — credential prompt + orphan gitlink (TMX-086/TMX-087) → 2026-09-01

Status: DONE (both fixes landed + verified; release documentation authored this commit). Full-suite retest UNCONFIRMED: — see below.

Trigger (verbatim operator report, 2026-09-01): "We cannot install / setup tmx (tmux) on the System! Running our bash script gets stuck asking for GitHub credentials, which MUST NOTEVER happen! We shall rely only on ssh key (git protocol)!" Investigated via /

superpowers:systematic-debugging per §11.4.102. Universal §11.4.238 applies — the defect was found by the OPERATOR attempting a real install, not by any automated gate this project ships; test 90 closes that coverage escape.

Root cause (TWO INDEPENDENT defects behind ONE symptom — either alone blocks the install, so fixing only one would have left the operator still broken; §11.4.194 exhaustive-review discipline is what surfaced the second):

- **TMX-086 — the credential prompt.** scripts/install.sh injected `-c url.https://github.com/.insteadOf=git@github.com: BY DEFAULT`. Git propagates `-c` flags to child processes via `GIT_CONFIG_PARAMETERS`, so the rewrite did NOT stay scoped to the top-level clone — it reached every RECURSIVE submodule clone. The constitution submodule nests submodules/helix_perf_cache -> `git@github.com:HelixDevelopment/helix_perf_cache.git`, the ONLY repo of the nine in the tree that is not anonymously readable; rewritten to HTTPS it cannot be fetched without credentials, so GitHub asks for a Username, and with nothing setting `GIT_TERMINAL_PROMPT=0` git BLOCKED on that prompt indefinitely. Under `curl | bash` the prompt is unanswerable. Anonymous-HTTPS census (all git config bypassed, no credential helper on host): `PUBLIC = vasic-digital/{tmux,Containers,token_optimizer,session_orchestrator,continuum,anti_bluff,design-toolkit,docs_chain} + HelixDevelopment/HelixConstitution; PRIVATE = HelixDevelopment/helix_perf_cache` (the trigger).
- **TMX-087 — the orphan gitlink.** The constitution submodule's index carried TWO gitlinks at the identical SHA `efd2c3fb2f880aaf2baf7f4819ce28a1ce3609cb`: a bare design-toolkit with NO `.gitmodules` mapping, and the correctly-mapped submodules/design-toolkit. Git refuses to recurse into an unmapped gitlink, so every consumer's recursive clone aborted with fatal: No url found for submodule path 'constitution/design-toolkit' in `.gitmodules` / fatal: Failed to recurse into submodule path 'constitution'. Introduced when the submodule was relocated under submodules/ (constitution commit `e499ffb`) without removing the old index entry.

Fix:

- **TMX-086** (all in scripts/install.sh): `DEFAULT_REPO_URL` changed to `git@github.com:vasic-digital/tmux.git`; the `insteadOf` rewrite is OFF by default and now an explicit opt-in via

TMX_INSTALL_HTTPS_REWRITE=1 / --https-rewrite, with the legacy
 TMX_INSTALL_NO_HTTPS_REWRITE=1 / --no-https-rewrite still
 honoured as an opt-OUT that WINS (§11.4.122 — the capability is
 preserved, only its default changed); exported
 GIT_TERMINAL_PROMPT=0, GIT_ASKPASS="", SSH_ASKPASS="" so a missing
 credential fails fast instead of hanging; exported
 GIT_SSH_COMMAND="\${GIT_SSH_COMMAND:-ssh -o
 StrictHostKeyChecking=accept-new -o ConnectTimeout=20}" so a
 first-contact github host key cannot hang either. Honest boundary
 (§11.4.6): accept-new still REFUSES a CHANGED key, an ssh key
 passphrase prompt is deliberately still allowed, and an operator-set
 GIT_SSH_COMMAND is never overridden.

- **TMX-087:** constitution commit b9096ac removes ONLY the orphan
 gitlink — .gitmodules untouched, submodules/design-toolkit keeps
 the same pinned SHA, so no capability is dropped. Pushed fast-
 forward (6f24df9..b9096ac) to all SIX constitution upstreams
 (github, gitlab, gitflic, gitverse, vasic_digital_github,
 vasic_digital_gitlab), no force-push (§11.4.113). This repo's
 constitution pointer is bumped 6f24df9 → b9096ac.

Tests: new scripts/tests/

90_install_ssh_only_no_credential_prompt.sh (§11.4.115 RED/GREEN,
 RED_MODE default 0 per the TMX-085 convention). Method: a recording
 git shim placed first on PATH captures the REAL argv and the
 INHERITED GIT_TERMINAL_PROMPT, so the assertions are RUNTIME
 evidence, not a grep of the installer source (§11.4.226). Sub-checks: **A**
 default clone URL uses the SSH form; **B** no insteadOf rewrite in the real
 git argv by default; **C** git inherited GIT_TERMINAL_PROMPT=0; **D**
 TMX_INSTALL_HTTPS_REWRITE=1 still enables the rewrite (capability
 preserved). RED_MODE=1 reconstructs a pre-fix copy of install.sh and
 asserts the rewrite IS injected — proving the test is not blind. Three
 paired mutations in meta_test_false_positive_proof.sh: M-INSTALL-
 SSH-DEFAULT (expect FAIL: A), M-INSTALL-HTTPS-REWRITE-DEFAULT (expect
 FAIL: B), M-INSTALL-NO-PROMPT-GUARD (expect FAIL: C).

Measured verification:

- Controlled A/B (same repo, same host, same minute, ONE variable):
 SSH as declared → exit=0, 6adcfb2 refs/heads/main; with the
 installer's rewrite → exit=128,
 fatal: could not read Username for 'https://github.com'.
- Test 90 GREEN (RED_MODE=0, default) → PASS=4 FAIL=0 SKIP=0, exit 0.
 Test 90 RED (RED_MODE=1) → PASS=1 FAIL=0 SKIP=0, exit 0, defect
 reproduced on the pre-fix artifact.

- Live clone observation: with the orphan gitlink removed, Containers and constitution checked out over SSH with NO credential prompt.
- **UNCONFIRMED: full-suite run_all.sh retest** — still in flight when this entry was authored; its outcome is not known here and is NOT claimed (§11.4.6 / §11.4.40 remains owed before the tag is cut).

Test-instrument bug found and fixed during authoring (§11.4.1 FAIL-bluff / §11.4.201(7)(c)): test 90's first argv extractor keyed on ' clone' WITH A LEADING SPACE, which matched ONLY when the defect was present (the pre-fix installer inserts -c ... before clone), producing a false FAIL on correct code. Fixed at the instrument layer to a word-boundary match and documented inline in the test.

Placeholder owed (§11.4.6): Fixed.md \$L carries (this commit – MUST be backfilled with the real SHA on landing) for both items' tmux-side closure commit — the same never-backfilled-placeholder class already corrected for TMX-079 / TMX-082 / TMX-083. Backfill on landing.

Full detail + captured evidence: Fixed.md \$L (TMX-086, TMX-087); CHANGELOG.md v1.0.44.

§3.34 — TMX-080/TMX-081 deep-dive investigation (no fix landed; documentation-only) → 2026-08-13

Status: IN PROGRESS / Operator-facing next-step needed. No code changed this entry — scripts/tmx.template and scripts/tmx-recycler.sh are IDENTICAL to before this investigation (two fix attempts were made, both reverted after proving insufficient — see below). Only Issues.md (§H1/§H2) was updated with corrected, more complete findings.

Trigger: operator request "Investigate TMX-080 and TMX-081 next", continuing directly from the TMX-084/TMX-085 cycle.

TMX-080 (test 27 sub-check 18, run-shell cwd-record hook): the original 2026-08-10 diagnosis ("iteration 1 specifically, deterministically") is CORRECTED — 8 fresh instrumented runs show the failure lands on VARYING iterations (2, 3 — never iteration 1 in this cycle's sample), confirming it is genuinely intermittent. A real, confirmed race was found (the wrapper's kill-session verb calls systemctl --user stop with KillMode=control-group immediately after tmux kill-session, which can SIGTERM an in-flight run-shell-spawned async child — reproduced directly in isolation) and two fix attempts

were made (delaying the `systemctl stop` until the scope's cgroup is observed empty) — **both made the test's failure rate WORSE (100% reproducible, vs. ~20-25% before) rather than better, and were reverted per the Iron Law rather than left in place.** A SECOND, deeper, source-code-confirmed mechanism was then found: tmux's `#{pane_current_path}` (tmux/osdep-linux.c `osdep_get_cwd`) resolves via `tcgetpgrp(fd)` (the PTY's current foreground process group), NOT the shell's tracked PID — so if the pane's shell (oh-my-bash, confirmed in use via this operator's `.bashrc`) spawns prompt-render subprocesses, different queries moments apart can read different processes' `cwd`. This is NOT yet confirmed against a live failure (today's non-reproduction prevented that), but is the strongest lead.

TMX-081 (test 87 G4, quiet-phase settle): could NOT be reproduced today — 24/24 standalone runs (4 individual + a 20-run background batch, full per-try `nr_throttled` deltas instrumented) all settled cleanly, typically on the first 1 s window. Per §11.4.7 this does NOT close the item — today's host conditions differ from 2026-08-10's. The SAME oh-my-bash/tcgetpgrp mechanism found for TMX-080 is offered as a shared-cause hypothesis (oh-my-bash's own CPU usage during prompt-render, under G4's tight quota, could extend past the 12 s settle bound) but is likewise unconfirmed against a live failure.

Recommended next step (not yet done): an experiment disabling oh-my-bash for test-created sessions specifically, to see if BOTH TMX-080's stale-path reads and TMX-081's settle timeouts stop reproducing — this would confirm or refute the shared-cause hypothesis directly. Needs either genuine host load, a cold oh-my-bash cache, or an artificially-injected slow prompt hook to force a live reproduction, since today's host conditions did not naturally produce one for either issue despite substantial effort (8 + 24 = 32 total instrumented runs across both).

Full detail + captured evidence: `Issues.md` §H1 (TMX-080) + §H2 (TMX-081), both updated 2026-08-13.

§3.33 — v1.0.43: verification-harness false-result classification (TMX-084/TMX-085) → 2026-08-13

Status: DONE (both fixes landed + verified, release-tag publish in progress this commit).

Trigger: operator-reported `install.sh` verification RED (SUMMARY: PASS=66 FAIL=5 SKIP=16, failing tests 27_state_persistence.sh 56_real_mouse_drag_copy.sh 57_reload_select_copy_paste.sh 59_oomd_preference_avoid.sh 87_server_scope_split.sh), investigated via `/superpowers:systematic-debugging`.

Root cause (two DISTINCT, unrelated defects, both in the verification HARNESS, neither in the wrapper):

- **TMX-084:** `scripts/tests/run_all.sh` classified each test file's verdict via a bare `grep -qE '^FAIL'|'^SKIP'|'^PASS'` (no delimiter check). Tests 56/57 print an honest SKIP-layer: `<reason>` informational note for one optional GUI-only sub-check while genuinely PASSing overall (real drag/copy/paste evidence) — the bare-prefix regex misread "SKIP-layer:" as a top-level SKIP verdict. The same carrier shape exists unfixed in many other tests' own internal "PASS=\$PASS FAIL=\$FAIL SKIP=\$SKIP" counters (a healthy FAIL=0 line would ALSO mis-fire).
- **TMX-085:** `scripts/tests/59_oomd_preference_avoid.sh` (the TMX-083 regression guard) defaulted to `RED_MODE=1` (reproduce-the-historical-defect polarity) — since `run_all.sh/verify.sh` never override `RED_MODE`, a bare invocation FAILED 100% deterministically against the genuinely-fixed wrapper, the opposite of what a standing regression guard needs.

27 and 87 were independently re-confirmed as the already-tracked TMX-080/TMX-081 (Issues.md §H, unchanged, Status still Queued) — no new action.

Fix: TMX-084 — extracted classification into a new, independently-tested `scripts/tests/lib/classify_verdict.sh` requiring a genuine verdict-line delimiter (`:`, `,`, or `()`) after the keyword, matching every real verdict line already used across the suite. TMX-085 — flipped test 59's default to `RED_MODE="${RED_MODE:-0}"`, matching the already-working convention 7 of the suite's other 11 `RED_MODE`-polarity tests already use.

Tests: new test 89 (`89_classify_verdict_carrier.sh`, §11.4.115 RED/GREEN, 6 fixtures — 3 carrier, 3 non-carrier); paired mutations M-CLASSIFY-CARRIER + M-RED-DEFAULT-59 in `meta_test_false_positive_proof.sh`.

Also backfilled: TMX-083's Fixed.md §J1 Closure commit: placeholder (was (this commit), now 6f9eae1) — the same never-backfilled-placeholder class already fixed for TMX-079/TMX-082, discovered while investigating this cycle.

§12.10 note: this document had gone stale across the ENTIRE TMX-083/v1.0.41/v1.0.42 cycle (the autonomous process that landed commits 76f3433/6f9eae1/92ef3a0/b29b9e1 never updated it) — backfilled retroactively in §3.32 below so the resumption record is complete.

Next in this commit: version tag `tmux-1.0.43` published to GitHub + GitLab with changelog.

§3.32 — v1.0.41 / v1.0.42: install-verification-RED fix + systemd-oombd victim-avoidance (TMX-083) → 2026-08-12 (retroactively documented 2026-08-13)

Status: DONE (released as `tmux-1.0.41` then `tmux-1.0.42` on GitHub + GitLab; this entry backfills the §12.10 gap left when those commits landed without a CONTINUATION.md update).

v1.0.41 — install-verification-RED fix, 4 distinct root causes across tests 22/58/77/86/88: (1) ambient `TMX_SERVER_SPLIT=1` from this host's `.bashrc/.zshrc` decided 3 test verdicts on provably-healthy code (fixed via new scripts/tests/lib/hermetic_env.sh, neutralising ambient `TMX_*` knobs at test entry — sourced by 58/77/86/88); (2) test 77 matched a 2-char plaintext sentinel as a bare substring over the whole pane buffer (scoped to the password-prompt line, §11.4.201(7)(a)); (3) test 58 raced a fixed 1.0s sleep against a slower first-prompt render (replaced with condition-based polling); (4) `~/.config/opencode/opencode.json` genuinely still lacked the codegraph MCP entry — the v1.0.40 CHANGELOG's claim that this was already fixed was a **documentation-layer bluff** (the file's mtime proved the write never landed) — now actually applied + verified.

v1.0.42 — TMX-083, systemd-oombd victim-avoidance: operator-reported (on the boba consumer project) "sessions get killed ... on strong hardware" despite the full TMX-079 series landing `MemoryMax=infinity/TasksMax=infinity`. Systematic-debugging refuted every TMX-079-series killer and surfaced the residual: `systemd-oombd.service` operates ORTHOGONALLY to `cgroup Max=` limits (PSI-pressure + swap-usage victim selection, not scope ceiling) — under a real memory-pressure spike it SIGKILLS the whole `tmx-<NAME>.scope`. Fix: `-p ManagedOOMPreference=avoid` added to all 3 scope/slice creation sites in `scripts/tmx.template`, version-guarded `sd_ver >= 249`. New test

59 (59_oomd_preference_avoid.sh, §11.4.115 RED/GREEN). Renumbered from v1.0.41 to v1.0.42 per §11.4.113 merge-onto-latest-main (a concurrent main branch had already shipped v1.0.41 while this work was in flight).

Full detail + captured evidence: Fixed.md §J (TMX-083);
CHANGELOG.md v1.0.41 + v1.0.42.

§3.31 — v1.0.40: split-topology test reconciliation (TMX-082) → 2026-08-11

Status: DONE (fix landed 74b35bd, real-live-session verified, released as tmux-1.0.40 on GitHub + GitLab).

Trigger: operator-reported install.sh verification RED (SUMMARY: PASS=68 FAIL=3 SKIP=15, failing tests 09_crash_isolation_scope.sh 15_per_session_cgroup_distinct.sh 22_codegraph_mcp_wired.sh), investigated via /superpowers:systematic-debugging.

Root cause: new split-topology crash-isolation test coverage (test 09's T7, test 15's T7-T9) landed on main via a separate work stream (adf46d5, "Auto-commit", 2026-08-11) one day after v1.0.39 shipped, without the TMX_CPU=auto opt-in the v1.0.39 default change requires to engage the split topology — the same class of gap already reconciled in test 87's G6/G8 during the v1.0.39 review cycle, not yet propagated to this independently-landed coverage. Full detail + captured evidence: Fixed.md §I (TMX-082).

Fix: added TMX_CPU=auto to the four affected TMX_SERVER_SPLIT=1 spawns in tests 09 and 15. Also fixed test 22 (host-local opencode.json MCP wiring gap, this host only, not tracked in git).

Real-session verification (this cycle): install.sh clean install, then a real tmx new -s <name> -d session created directly via the installed CLI with TMX_SERVER_SPLIT=1 TMX_CPU=auto — root-verified via /proc/<pid>/cgroup (kernel ground truth, not a tool's exit status) that the pane's actual shell lands under tmxw.slice/tmxw-<name>.slice/. An initial systemctl is-active check against an unescaped unit name had misreported "inactive"; resolved as a verification-command bug (missing \x2d hyphen-escaping), not a wrapper defect — confirmed by cross-checking systemd's identical escaping behaviour on unrelated, known-good device units on the same host.

Also confirmed, during this investigation, that the operator's 3 pre-existing real sessions (atmosphere-4224, btop-0626, tmx-7268) had already died — journal timestamps show their scopes started and (for one) stopped entirely on 2026-08-10, hours before this cycle's `install.sh` run, with no reboot since (`uptime -s = 2026-08-10 11:33:12`) and no OOM-killer/recycler/crontab activity found in that window. Pre-existing/unrelated to this cycle's work, not a regression it introduced.

Next in this commit: version tag `tmux-1.0.40` published to GitHub + GitLab with changelog.

§3.30 — v1.0.39: no resource or lifetime limit by default (TMX-079) → 2026-08-10

Status: DONE (fix + tests + docs landed, released as `tmux-1.0.39` on GitHub + GitLab).

Trigger (verbatim operator report, 2026-08-10): "sessions and processes get killed and wiped out ... on powerful hardware with enough resources — it MUST NOT happen".

Root cause (systematic-debugging): three unconditional-by-default caps in `scripts/tmx.template`: (1) the idle-timeout recycler (`scripts/tmx-recycler.sh`) tore down ANY session with no client attached for ≥ 900 s regardless of activity — the dominant cause, and the ONLY mechanism that actually kills an already-running session's processes irrespective of host capacity; (2) CPUQuota applied unconditionally (host-adaptive since v1.0.37, no off switch); (3) a hardcoded `TasksMax=4096` with no override knob at all. Memory was already correctly "fully elastic" by default — CPU/tasks/lifetime brought into that same model.

Fix: `TMX_CPU/TMX_TASKS` unset $\rightarrow$ unlimited (`=auto` opts in to the prior host-adaptive/legacy-fixed value); `TMX_RECYCLE_IDLE_SECS` unset/0 $\rightarrow$ never auto-recycled; Darwin `TMX_CPU_HARD_SEC/TMX_PROC_MAX` $\rightarrow$ unlimited by default. Full detail + captured evidence: `Fixed.md` §H (TMX-079).

Tests: new test 88 (§11.4.115 RED/GREEN, live cgroup + `tmux-hook` readback); tests 15/86/87 reconciled to the new default (§11.4.120 — a correct fix legitimately breaking a stale gate assumption); meta-test `M-NOLIMITS` added, `M-CPUADAPT` fixed (was silently escaping post-fix).

Discovered + tracked, NOT part of this fix (confirmed pre-existing via baseline A/B, §11.4.114): TMX-080 (test 27 hook-record timing, Issues.md §H1), TMX-081 (test 87 G4 quiet-phase settle heuristic, Issues.md §H2).

Next in this commit: independent code review (dispatched, Fable model per §11.4.209), full non-destructive `run_all.sh` retest (in progress), `meta_test_false_positive_proof.sh` full sweep, then `commit_all.sh` + version tag `tmux-1.0.39` published to GitHub + GitLab with changelog.

§3.28 — Wizard + session-password redesign (2026-07-05) — docs landed, code in sibling tasks

Status: IN PROGRESS — Task 11 of 14 (documentation) landing this commit.

Plan: `docs/superpowers/plans/2026-07-05-tmx-wizard-password-redesign.md` **Spec:** `docs/superpowers/specs/2026-07-05-tmx-wizard-password-redesign-design.md`

Four user-visible requirements (operator mandate 2026-07-05), tracked as new OPEN workable items in the SQLite SSoT + Issues.md §G:

- **TMX-072** (Feature) — wizard-created sessions get a random 4-digit name suffix (name-NNNN; `TMX_EXACT_NAME=1` opts out).
- **TMX-073** (Feature) — password input masked with `*` while typing.
- **TMX-074** (Bug) — reopening a password-protected (idle-recycled) session no longer asks for the password twice; verify-once.
- **TMX-075** (Feature) — wizard offers a numbered picker of existing sessions (+ 0) None) on blank input.

This task (Task 11 — documentation) delivered: README What's-new + docs-map rows; `docs/guides/tmx-shell-integration.md` wizard section rewritten (suffix, picker, masking, `TMX_EXACT_NAME`); new `docs/guides/tmx-session-passwords.md` standalone guide + Mermaid decision-flow diagram (rendered PNG, confirmed image-not-raw-source in the PDF per §11.4.168); new `docs/guides/FAQ.md`; the 4 workable items added to the SQLite SSoT + Issues.md §G regenerated byte-identical (`diff` clean); all touched Markdown re-exported to HTML/PDF/DOCX.

Next (remaining plan tasks, concurrent): the CODE tasks — has-password Go subcommand, `_read_password_masked` + restructured new/attach in `scripts/tmx.template`, the always-create + picker wizard in

scripts/tmx-shell-init.sh.template, and tests 77–84 + 66/68 reconciliation — then full-suite retest + a validated release tag per §11.4.40 / §11.4.126. Terminal goal: a new fully-validated version published to all remotes.

Note (§11.4.6): workable-items validate reports a single pre-existing §11.4.54 ticket sequence gap: expected TMX-071, found TMX-072 — TMX-071 was a previously-reverted item and the next_atm_id counter was already at 72, so the new items correctly start there. NOT to be back-filled or renumbered.

§3.27 — v1.0.26 RELEASED (per-session color) + BOTH HOSTS INSTALLED → 2026-06-19

Status: RELEASED v1.0.26 / versionCode 27 — tag v1.0.26 at 5244f4e (13ef83b), published GitHub (<https://github.com/vasic-digital/tmux/releases/tag/v1.0.26>) + GitLab (<https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.26>). main HEAD 5244f4e identical on local + github + gitlab.

Feature: operator-chosen per-session tmux color via `tmx new -s name:color[:ignored]`. Color validated (tmux names / colour0-255 / #hex), applied to all 4 "green" surfaces (status-bar bg, active-pane border, clock-mode-colour, current-window bg), persisted in `~/.tmx/state.json` (schema 1→2 additive color field), re-used on bare-name re-runs. Precedence: inline > persisted > hostname > default-green. Escaped \: in name; extra :fields ignored. `tmx-state-bin v1.0.9` → `v1.1.0` (set-color/get-color + list 4th col). Invalid color rejected (exit 5) before any session created. macOS bridge unchanged → §11.4.81 parity by construction.

Anti-bluff evidence (§11.4.5/§11.4.69):

- scripts/tests/63_session_color.sh — 8/8 PASS, every assertion reads LIVE show-options from the real server. Deterministic 3× (§11.4.50).
- scripts/tests/64_session_color_parse_unit.sh — 17/17 pure parser/validation unit tests incl. bash↔Go canonical-name-list parity.
- §1.1 paired mutations M25 (strip _apply_color) + M26 (neutralize set-color) both CAUGHT by test 63.
- HelixQA Challenge TMUX-CH-53 (blocker).
- **Mistborn INSTALLED** (setup.sh --rebuild): native Darwin arm64 build, tmux 3.6a, tmx-state-bin v1.1.0, VERSION 1.0.26. Verify gate PASS=57 FAIL=0 SKIP=6 (test 63 = 8/8, test 64 = 17/17 ran during

install). **Live end-user proof on the installed build:** `tmx new -s proofofwork:red → status-style=bg=red, clock-mode-colour=red, pane-active-border-style=fg=red; persisted (get-color → red) + bare re-run reuses it; #hex works (bg=#e63946); invalid notacolor rejected exit 5 + no server.`

**** nezha — INSTALLED**** (/home/milosvasic/Projects/tmux, 2026-06-19): after the operator confirmed nezha back online (§11.4.144 unblock condition met — `ssh ... nezha.local` returned `Linux x86_64`), clean fast-forward 05569e5 (v1.0.25) → e1a2530 (v1.0.26) (local in-flight D2 TMPDIR edits stashed — already superseded upstream at 5f86936), `setup.sh --rebuild` → `Linux ELF tmux 3.6a + tmx-state-bin v1.1.0`. Verify gate **PASS=49 FAIL=0 SKIP=14**; test 63 **8/8** (T8 hostname fallback → `bg=colour94`, matching Mistborn → cross-OS parity §11.4.81) + test 64 **17/17**. Live color proof on nezha: `tmx new -s nezproof:red` → all 4 surfaces red; persisted; #hex; invalid rejected exit 5. NEZHA-INSTALL-v1.0.26-001 closed → Fixed.md. **Both hosts now at v1.0.26, clean + verified.**

§11.4.151 release-prefix — NOT adopted this release: project tag history is v1.0.6...v1.0.25 (no prefix); operator chose v1.0.26 to keep the series continuous (decision recorded). The universal `tmux-<ver>` prefix remains a standing operator decision for a future major release boundary.

Known / tracked separately (not regressions): M24-ESCAPE-001 NOW FIXED (v1.0.27, meta-test CAUGHT 37/0/17). Test 13 Darwin `ulimit -u` honest-gap (§11.4.81, pre-existing).

§3.28 — v1.0.27 RELEASED: name:color prompt rejection fixed + Issues.md burn-down (M24/A2/D2 all closed) → 2026-06-19

Status: RELEASED v1.0.27 / versionCode 28 — tag v1.0.27 pushed + published GitHub (<https://github.com/vasic-digital/tmux/releases/tag/v1.0.27>) + GitLab (<https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.27>). Both hosts validated:

- Mistborn (macOS arm64): 58 PASS / 0 FAIL / 6 SKIP + meta 37 CAUGHT / 0 ESCAPED / 17 SKIPPED
- Nezha (Linux x86_64): 49 PASS / 0 FAIL / 14 SKIP (meta pending post-push pull)
- commit 67d5f01 at HEAD + tag v1.0.27 — github + gitlab identical.

Operator report (2026-06-19) — §11.4.138 operator-escape:

```
[tmx] invalid session name 'home:red'; allowed: [A-Za-z0-9_.-]
{1,64}
```

ROOT CAUSE (Phase 1 complete, fix applied). The tmx wrapper's `_parse_session_value()` already handled `NAME:color` correctly — the gap was the shell-init prompt AND SSH-dispatcher running BEFORE the wrapper with `[A-Za-z0-9_.-]` char-set + 64-char limit (a textbook §11.4.108 SOURCE→USER-VISIBLE gap: GREEN suite, broken-for-user feature). Test 65 existed for this exact bug (written as the anti-bluff regression guard for v1.0.26) but the underlying templates had not been updated.

Fix applied (all 4 changes reconciled per §11.4.120):

- `scripts/tmx-shell-init.sh.template` — char-set `[A-Za-z0-9_#.-]`, max 64→80
- `scripts/tmx-ssh-dispatch.sh.template` — same fix
- `scripts/tests/65_shell_init_color_prompt.sh` — updated T2/T3 for full-spec forwarding (not pre-parsed)
- `scripts/tests/35_session_name_validation.sh` — LONG65→LONG81, limit 64→80

Verification: 58 PASS / 0 FAIL / 6 SKIP (63=8/8, 64=17/17, 65=6/6). Meta 35/35 caught.

§3.28.1 — M24-ESCAPE-001 (Stream 1) → DONE. Root cause: M24 mutation used `re.subn(pat, '', src, count=1)` which matched `_apply_color()` first (wrong function), while test 26 exercises `_apply_host_color()`. Fixed by removing `count=1` → strips from BOTH functions. Meta-test before: 34 CAUGHT / 3 ESCAPED (M24 was 1 of 3). After: **37 CAUGHT / 0 ESCAPED / 17 SKIPPED**. M24 verified: stripped 6 set-lines across both color functions → test 26 FAILs (T2/T3/T4), reverted → PASSes.

§3.28.2 — A2 RUNALL-NATIVE-RESOLVE-001 (Stream 2) → DONE. `run_all.sh` now OS-aware: builds `TMUX_BIN_DEFAULT="$REPO_ROOT/tmux/build-darwin/bin/tmux"` with fallback to `tmux/build/bin/tmux`, respects `$TMUX_BIN` override. Gate CM-RUNALL-OS-AWARE PASSes.

§3.28.3 — D2 TMPDIR-HARDCODE-001 (Stream 3) → DONE. 35 test files updated: bare `/tmp` paths replaced with `"$SCRATCH/..."` where `SCRATCH="${TMPDIR:-/tmp}"`, plus writability preflight guard per §11.4.3. Gate CM-NO-HARDCODED-TMP-SCRATCH PASSes (16 cwd/state/dispatch tests).

QA: docs/qa/2026-06-19-v1.0.27-burndown/.

Anti-bluff verification (current host — macOS Mistborn):

- Full suite: **58 PASS / 0 FAIL / 6 SKIP** (test 63=8/8, 64=17/17, 65=6/6)
- Meta-test: **37 CAUGHT / 0 ESCAPED / 17 SKIPPED**
- bash scripts/setup.sh — GREEN, installed.
- bash scripts/setup.sh --verify-only — GREEN.

Nezha deploy: completed 2026-06-19 post-tag (verified 49/0/14 + meta 37/0/17).

§3.29 — v1.0.28 RELEASED: per-session password protection + governance sync (§11.4.159–170) → 2026-06-28

Status: RELEASED v1.0.28 / versionCode 29 — tag v1.0.28 at f30cd8b, published GitHub (<https://github.com/vasic-digital/tmux/releases/tag/v1.0.28>) + GitLab (<https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.28>).

What landed:

- **Per-session password protection** — tmx new -s NAME prompts for optional password; tmx attach -t NAME verifies before granting access. SHA-256 hashed storage, schema v3, graceful degradation.
- **tmx-state-bin v1.2.0** — new set-password / verify-password subcommands.
- **Test 66** — 5 invariants × 3 iterations (15/15 PASS).
- **Meta-test M27** — verifyPassword always-true mutation CAUGHT by test 66 T3.
- **Governance sync** — constitution 1d408cb → 1576d3d; §11.4.159–§11.4.170 propagated to all 5 governance carriers.

Verification: Full suite PASS=51/FAIL=0/SKIP=14. Meta-test 55 CAUGHT/0 ESCAPED/8 SKIPPED.

Pending: Release tag, deploy on Mistborn + mistborn.local.

§3.26 — v1.0.25 RELEASED (test fixes + submodule currency + governance cascade) → 2026-06-16

Status: RELEASED v1.0.25 / versionCode 26 — tag at 3dd6a7c, published GitHub (<https://github.com/vasic-digital/tmux/releases/tag/v1.0.25>) + GitLab (<https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.25>). Both hosts (macOS Mistborn

- nezha) synced to 05569e5 (= tag + governance cascade), VERSION 1.0.25, tmux 3.6a, setup + smoke (tmx new/ls/kill) GREEN. Authoritative validation: nezha rel25c destructive gate verify PASS=50 FAIL=0 SKIP=11 (V=0) + meta 52 CAUGHT/0 ESCAPED; macOS setup run_all 55/0/6. Test 17 determinism: macOS 5/5 + nezha 20/20 under full all-core CPU saturation. **Governance cascade (05569e5):** CLAUDE/AGENTS/QWEN appended §11.4.101–158 verbatim + **GEMINI.md created** (§11.4.157 lockstep, byte-identical to CLAUDE.md); §11.4.65 siblings regenerated. **Operator-pending:** §11.4.151 release-prefix (tmux-<ver>) — NOT adopted mid-series (project uses vN.N.N); decide before next release. Both hosts at latest, clean.

(Historical detail of the cycle follows.) Per operator "always target latest of all submodules": advanced HelixConstitution 3f4d690 → 1d408cb (+67 commits, §11.4.140–158); Containers already at latest 18ed03d. Functional inheritance verified (test 18 PASS=10/0); release gate (verify.sh §11.4.87..99) unaffected. Host-level: codegraph macOS skew fixed (stale ~/.local/bin symlink → both hosts 1.0.1); 668 dead macOS test sockets removed (678→10, no live session touched).

Test 17 flake — root-caused + fixed (A49, 2 vectors).

17_scrollback_copy_mode.sh flaked under heavy host load (§11.4.1 FAIL-bluffs, NOT a product defect — scrollbar works for the end user). Vector 1: the T4 generator spelled SCROLLMARK_FIRST/SCROLLMARK_LAST inline, so the terminal's echo of the TYPED command line contained SCROLLMARK_LAST and the GEN_OK poll matched the command echo before any real output (forced-slow probe on nezha: command-echo grep=1, real-output grep=0). Fixed by assembling markers from a shell variable (M=SCROLLMARK; echo \${M}_LAST) → every grep matches REAL OUTPUT only. Vector 2: under all-core CPU saturation a capture-pane grid dump can momentarily lack the oldest FIRST marker although the buffer genuinely retains all 3000 lines (captured: 3006 lines, LAST present, 2998 numbered markers, history-limit 50000). Fixed by asserting retention via the **race-free #{history_size} counter** (>= 2900, display-message -p) at T4.2, not a grid dump (the no-eviction value is a deterministic ~2982; the OLD 2000 cap pins it at 2000 and evicts line 1)

— T4.3/T4.4 (operator copy-mode reach of FIRST) already proved retention robustly and never raced. Test source only (scripts/tests/17_scrollback_copy_mode.sh); assertions of the end-user guarantee unchanged. **Determinism (§11.4.50) — PROVEN:** macOS N=5 GREEN (5/5) + nezha N=20 under full all-core CPU saturation GREEN (20/20); history_size identical every run (macOS 2981 / nezha 2982 — counter does not race). Pending before tag: the §11.4.40 full destructive gate (rel25c, running) GREEN.

Known follow-up (non-blocking): verbatim/cascade mirroring of §11.4.140–158 into project CLAUDE.md/AGENTS.md/QWEN.md (for non-@import agents) — functional inheritance already works via the import; release gate does not enforce these. Operator-pending on nezha: sudo bash scripts/build_oom_set.sh --install + sudo apt-get install -y stress-ng to promote tests 08/12/14 from SKIP to PASS.

§3.25 — v1.0.24 RELEASE (distribution orchestrator) → 2026-06-16

Status: RELEASING v1.0.24 / versionCode 25. Tooling + library-hardening release (tmux product surface unchanged from v1.0.23). Bundles §3.24 (tmx-orchestrator + Containers fixes) under a tagged version published to GitHub + GitLab. Release gate green: nezha verify.sh PASS=50/0/11 + meta 52 CAUGHT/0 ESCAPED; nezha Containers full suite (unit + integration vs real podman) all 40 pkgs ok; macOS run_all 55/0/6; adversarial code-review GO. Containers pointer 61e01dc → 18ed03d (pushed github+gitlab).

§3.24 — Distribution orchestrator binary (Containers submodule) + nezha as heavy-test host → 2026-06-16

Status: DONE — orchestrator working, validated on nezha, committed.

What landed (A48). Operator mandate: "Make a proper binary using the Containers submodule lib to orchestrate the distribution — for when/if we need it for our testing needs." Delivered scripts/tmx-orchestrator/ (Go consumer of the decoupled

digital.vasic.containers submodule via replace): hosts / distribute / down. nezha.local registered in Containers/.env (gitignored). Extended the Containers library (§11.4.76) with ContainerRequirements.Ports

- -p rendering (1b9da9b) and a command-injection allowlist fix from the automated security review (82bd586) — both pushed github+gitlab; tmux pointer 61e01dc → 82bd586.

Validation (captured — docs/qa/2026-06-16-tmx-orchestrator/evidence.md): macOS conductor → nezha over SSH+podman: hosts real /proc resources; distribute → real nginx on nezha 0.0.0.0:18080->80/tcp, HTTP health 200 (podman ps + curl confirm); down → REMOVED-CLEAN. Runtime test scripts/tests/62_distribution_orchestrator.sh PASS=4/0 (opt-in TMX_TEST_REMOTE=1). Containers full unit suite on nezha all 37 pkgs ok. Two real bluff-audits (feature + security guard). Containers submodule was NOT modified for tmux-specifics (CONST-051).

Post-landing validation + hardening (2026-06-16, parallel-stream round): Dual-host green — macOS full suite PASS=55/0/6 (zero regression; +1 SKIP = new test 62), nezha Containers integration tests (-tags=integration) all 40 pkgs ok against real podman. Orchestrator hardened: distribute now prints a cleanup hint when the health check fails (container left running for inspection, no silent leak — verified via a real induced HTTP failure). README documents the --health tcp vs http distinction (TCP connect-only false-passes on a published port; HTTP forwards through — verified). §11.4.65 .html/.pdf siblings generated for the new/changed docs. nezha synced to tmux a130a8f + Containers 82bd586.

Adversarial code-review round (§11.4.125/§11.4.134 review-until-GO, 2026-06-16). A dispatched review found 2 major + minors; all fixed with real evidence:

- **MAJOR (lib socket leak):** SSHExecutor.Execute released the pooled ControlMaster ref only on success → leaked on error paths. Fixed (defer release on all paths) in Containers a856865 + white-box integration regression test pkg/remote/execute_release_test.go — RED (refs=1) → GREEN (refs=0) proven on nezha.
- **MAJOR (down FAIL-bluff):** cmdDown printed "teardown complete" + exit 0 even when a host was unreachable. Fixed — tracks per-host failures, exit 1 when teardown can't be confirmed. Proven: unreachable host → exit 1; reachable → exit 0.
- **minors:** health-check timeout/cancel now distinguished from a real failure; test 62 free-port probe uses portable ss -ltn (not bash-

only /dev/tcp). Re-review verdict: **GO — no blocking findings** (all 4 prior fixes confirmed sound, no regressions, white-box test verified not-false-PASS). 2 minor notes also closed: ExecuteStream same-class ref leak on rare cmd.Start failure (Containers 18ed03d) + orchestrator --port/--publish range-validation. tmux Containers pointer → 18ed03d.

Open follow-ups: none. The binary is on-demand ("when and if we need it").

§3.23 — libtinfo cross-distro static-link fix + test-determinism hardening → v1.0.23 (2026-06-16)

Status: DONE — RELEASED v1.0.23 (2026-06-16). Dual-host re-validated: nezha.local (ALT Linux) synced to v1.0.22 baseline, setup.sh run, full destructive suite executed on real Linux hardware.

What landed. (1) **libtinfo fix (A46):** the containerized Linux build linked libtinfo.so.6 dynamically; on ALT Linux (no NCURSES version nodes) the loader warned on every invocation. Fixed by static terminfo link (LIBTINFO_LIBS="-l:libtinfo.a" in docker/build_inside_container.sh) — ldd shows no libtinfo, jemalloc/glibc stay dynamic, warning gone. New test 61 + verify gate CM-NO-DYNAMIC-LIBTINFO. (2) **Test-determinism (A47):** test 27 (macOS cd-poll), tests 12/14 (Linux OOM-poll) — §11.4.1 timing races fixed, now deterministic. (3) macOS test 22 codegraph OpenCode wiring repaired (host ~/.config/opencode/opencode.json, §11.4.78).

Validation (captured — docs/qa/2026-06-16-libtinfo-crossdistro/evidence.md): nezha verify.sh EXIT 0 PASS=49/0/11 (tests 12/14 PASS w/ real kernel OOM-kill, deterministic 5/5); meta-test 52 CAUGHT / 0 ESCAPED; macOS run_all PASS=55/0/5; test 61 PASS; libtinfo warning 0 lines (was 1/invoke).

No open follow-ups. Issues.md zero non-terminal items. Honest SKIPS remain: 08_oom_score_adj (root/setcap), GUI/clipboard/remote/DB opts.

§3.22 — Apple container integration: on-demand containerized Linux under macOS for testing → v1.0.22 (2026-06-13)

Status: DONE — RELEASED v1.0.22 (2026-06-13). Testing-capability addition (no tmux source/wrapper change — shipped binary + tmx wrapper unchanged from v1.0.21).

Apple container 1.0.0 incorporated into the vasic-digital/Containers submodule as a new generic pkg/crossbuild/apple_container.go backend + RunInLinuxContainer (unit + integration tests + challenge). Proven on this macOS 15.5/arm64 host: real container run returns Linux aarch64 (host is Darwin), host-dir mount round-trips, paired mutation (strip --mount → exit 99). Consumed extend-don't-reimplement per §11.4.74 / §11.4.76.

tmx harness scripts/test_apple_container.sh builds tmux 3.6a INSIDE an Apple-container Linux VM (genuine Linux ELF: osdep-linux.o + libjemalloc.so.2 + libevent_core) and runs the suite. **Evidence** docs/qa/2026-06-13-apple-container/: uname.txt=Linux aarch64; tmux-version.txt=tmux 3.6a; elf-proof.txt (ELF magic + jemalloc ldd); build.log; run_all.log + summary.txt = **PASS=30 / FAIL=0 / SKIP=28** (base image ubuntu:22.04). 28 SKIPS are honest §11.4.3/§11.4.81 topology gaps (no systemd-cgroup, no physical terminal/clipboard/mouse, no host tooling); the 30 core tmux tests RAN + PASSED. Release notes for GitHub + GitLab at docs/qa/2026-06-13-apple-container/RELEASE_NOTES.md. Fixed.md A44; CHANGELOG v1.0.22 (Sources verified 2026-06-13 against apple.github.io/container/ documentation/); VERSION 1.0.22/23.

§3.21 — Copy/paste ARCHITECTURE fix (mouse off default) + HelixCode crash forensics → v1.0.21 (2026-06-13)

Status: DONE — RELEASED v1.0.21 (2026-06-13), committed b905c9b pushed to github + gitlab. Full validation captured: setup verify gate PASS=53/FAIL=0/SKIP=5 (~/.tmux.conf installed); test 59 wire-level RED→GREEN; test 57 determinism 10/10 clean + 8/8 hostile-clipboard (0 FAIL); collateral sweep 41 PASS/0 FAIL ×2; workable-items DB 14/14 Go tests + byte-identical round-trip; meta sweep 51 CAUGHT / 0 ESCAPED / 8 SKIP (M-MOUSEDEFAULT caught). QA evidence: docs/qa/2026-06-13-mouse-off-default/, docs/qa/2026-06-13-server-death/, docs/qa/

2026-06-13-helixcode-crash/. Bug 1 (HelixCode crash) → Issues.md F1
Operator-blocked (operator runs docs/qa/2026-06-13-helixcode-crash/
diagnose.sh).

Bug 2 (copy/paste) — DONE at source, anti-bluff proven. Operator (2026-05-28 .. 2026-06-13, all emulators): "select multiple lines does not work ... must work Linux+macOS ... must scroll and always be selectable — right-click → Copy." Systematic-debugging: 3 prior binding-level fixes (v1.0.15/18/20) failed → question the architecture. ROOT CAUSE (wire-level, no guessing): set -g mouse on emits mouse-tracking DECSET enables (CSI ?1000h/?1002h/?1006h) that SUPPRESS the terminal's native selection + right-click→Copy. FIX: default flipped to set -g mouse off (terminal owns the mouse → native select/copy/scroll everywhere); tmux mouse on demand via prefix m. PROOF (test 59, real PTY): mouse on = 6 enable-DECSET, mouse off = 0; after prefix m = 3. 4-layer: verify.sh L1 mouse off gate + test 59 (RED→GREEN) + meta M-MOUSEDEFAULT + regression sweep (56/57/58 enable mouse for on-demand path; 17 + TMUX-CH-17 assert the default; 44/45/47/48 unaffected). Docs (scrolling/clipboard/guide §5.7) updated; Fixed.md A43; CHANGELOG v1.0.21; VERSION 1.0.21/22.

Bug 1 (HelixCode session crashes the whole terminal) — RESOLVED 2026-06-13 (operator-confirmed; Fixed.md A45) — was OPERATOR-BLOCKED (Issues.md F1).** Fresh tmx new -s HelixCode creates+attaches cleanly over a real PTY; 5 crash vectors (passthrough/extkeys/attach-reload/rename-format/stale socket) each reproduced headlessly + DISPROVEN as standalone causes. Crash needs operator-side runtime state (the live HelixCode TUI agent). Ready operator diagnostic + forensics: docs/qa/2026-06-13-helixcode-crash/.

NEXT (resume here): (1) build/sync workable-items DB; (2) sync_all_markdown_exports.sh (docs changed); (3) bash scripts/setup.sh (regenerate wrapper for this checkout — current installed wrapper points at a non-existent /Users/. . . path — + reinstall conf); (4) full retest run_all.sh + meta on the committed tree (conf mutations need clean tree); (5) commit_all.sh push to all upstreams.

§3.20 — tmux PASTE fix + tmx reload + attach-reload → v1.0.20 (2026-05-29)

Status: DONE — RELEASED v1.0.20 (2026-05-29). GitHub: <https://github.com/vasic-digital/tmux/releases/tag/v1.0.20> · GitLab: <https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.20>. Operator-confirmed (drag-select + Cmd-V pastes selected content). Dual-host validated at release commit 0205b44: Mistborn run_all 53/0/4 + meta 50/0; nezha

run_all 44/0/13 + test 58 PASS. Root cause of the long copy/paste saga: a tmux server loads config only at startup, so re-opened long-lived sessions kept stale bindings — fixed by `tmx attach auto-reload + tmx reload + plain-drag override + POSIX paste binding`. Tests 56/57/58 (real-mouse SGR + clipboard) + meta 50/0.

- **Root cause of "still can't copy"**: the live Herald session was started BEFORE the mouse fix and a tmux server loads config only at start → stale forwarding binding. Fix is reload-of-running-sessions. Added `tmx reload [-t NAME]` (source-file into running sessions; skips dead sockets).
- **PASTE FIX**: prefix P binding `tmux load-buffer - <<< "${#@clip-read})"` was broken 3 ways — `#{@clip-read}` nested-quote collision ("completely new value"), `<<<` bash-herestring fails under `/bin/sh` (dash/Linux), literal `\;`. Rewritten POSIX-clean (single-quoted run-shell wrapping inline double-quoted probe | `tmux load-buffer - && tmux paste-buffer -p`). Real prefix-P now pastes the EXACT clipboard value.
- **Tests**: NEW test 57 (A stale-repro / B reload-fixes-copy / C paste-buffer / D REAL prefix-P exact-value / E no-<<< POSIX guard), 3/3. test 46 T4 + verify.sh Layer-1 gate updated to accept the POSIX paste binding + ADDED gates for plain-drag override + prefix+m. Meta mutations M-PASTE + M-PLAINDRAG
 - M-MOUSETOGGLE. Evidence: docs/qa/2026-05-29-paste-fix-reload/.
- **Mistborn**: installed via setup.sh (verify GREEN), run_all 52/0/4, meta 49 CAUGHT / 0 ESCAPED. Copy + paste + reload all proven on the installed host.
- **HONEST NOTE**: during socket cleanup the live Herald tmux server ended (no tmux servers running afterward); exact cause undetermined (NOT guessed). Aggressive socket-file pruning around a live session was careless — stopped. Stale Herald socket removed.
- **PENDING (operator gate)**: operator to manually validate copy+paste in a FRESH `tmx new -s X` session; THEN nezha setup + validation; THEN next release.

§3.19 — Dual-host update + full test/verify + v1.0.19 release (2026-05-29)

Status: DONE — RELEASED v1.0.19 (2026-05-29). GitHub: <https://github.com/vasic-digital/tmux/releases/tag/v1.0.19> · GitLab: <https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.19>. §11.4.40 full retest at the release commit c5235f8: Mistborn run_all 51/0/4 + meta 48/0; nezha

run_all 43/0/12 + meta 48/0; M22 CAUGHT on both. v1.0.19 = the codegraph non-interactive PATH probe in run_all + meta-test (no product/runtime change). Both hosts verified at the release commit. Mistborn (Darwin): run_all PASS=51 FAIL=0 SKIP=4, meta 48 CAUGHT / 0 ESCAPED (M22 CAUGHT). nezha (Linux): run_all PASS=43 FAIL=0 SKIP=12, meta 48 CAUGHT / 0 ESCAPED (M22 CAUGHT). Fix landed this round: added the A31 npm-prefix PATH probe to scripts/tests/run_all.sh + meta_test_false_positive_proof.sh (commits 5d6196c, a0a3f85) so the codegraph tests (20/22) + M22 mutation resolve the CLI in NON-INTERACTIVE shells too — nezha's .bashrc adds ~/.npm-global/bin only behind an interactive-guard, so standalone run_all over ssh-batch previously failed 20/22 spuriously while real interactive agent usage worked. test 16 nezha full-suite flake ((window-name test, passes standalone)) did not recur this round.

§3.18 — v1.0.18 REAL mouse-copy fix: plain drag selects+copies in Claude Code, proven with a real mouse (2026-05-29)

Status: DONE — RELEASED v1.0.18 (2026-05-29). GitHub: <https://github.com/vasic-digital/tmux/releases/tag/v1.0.18> · GitLab: <https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.18>. Real-mouse PROVEN on BOTH platforms: Mistborn (suite 51/0/4 + meta 48/0 + real cliclick cursor drag over iTerm2 → pbpaste) AND nezha Linux (verify 43/0/12 GREEN + headless SGR real-mouse-drag → @clip sink, after the TERM=xterm-256color test-harness fix). Commits 5c667cc → 06e43d0 → 69d1418. NOTE: test 16 (window-name) flaked once under full-suite load on nezha (passes 3/3 standalone + GREEN on retry) — pre-existing §11.4.50 reliability issue, unrelated to the mouse fix; tracked for a future poll-with-budget fix.

Operator re-reported that select+copy still failed in Claude Code and demanded it work on BOTH Linux + macOS. The v1.0.17 prefix m toggle was insufficient (required a magic keystroke). Root cause reproduced with a REAL mouse: in a mouse-tracking app (mouse_any_flag=1) tmux's built-in root MouseDrag1Pane forwards the drag to the app → no selection. **Fix:** override bind -n MouseDrag1Pane if -F '#{pane_in_mode}' 'send -M' 'copy-mode -M' in scripts/tmux.conf.template — plain drag always enters copy-mode + copies on release; click+wheel still reach the app; no terminal-specific modifier (works Linux + macOS). PROVEN with a real mouse: headless SGR-1006 drag injection (mouse_any_flag=1) → @clip sink, regression-discriminating, 3/3 (test 56, cross-platform); + real cliclick cursor drag over iTerm2 → pbpaste (macOS GUI layer). Meta-test M-PLAINDRAG

CAUGHT; meta 48 caught / 0 escaped; suite 51/0/4. Commit 5c667cc.
Pending: dual-host setup.sh (refresh ~/.tmux.conf), nezha headless
test-56 proof, v1.0.18 release.

§3.17 — v1.0.17 two user-reported bug fixes + B3 closure + dual-host re-validation (2026-05-29)

Status: DONE — RELEASED v1.0.17 (2026-05-29). GitHub: <https://github.com/vasic-digital/tmux/releases/tag/v1.0.17> · GitLab: <https://gitlab.com/vasic-digital/tmux/-/releases/v1.0.17>. Dual-host setup GREEN (Mistborn suite 51/0/4 + meta 47/0; nezha 43/0/12, double-prompt real-HOME 2→1). CodeGraph 0.9.7 validate PASS both hosts. Commits 1de20c5 → a6a81df → bbb1467.

Operator reported two product defects 2026-05-29: (1) mouse select/copy unusable in tmx panes, especially under Claude Code; (2) double session-name prompt on new bash-login terminals. Plus directive to validate every fix with autonomous physical-proof tests, run setup.sh on Mistborn + nezha, document, and release via gh/ghlab.

- **A41 (double-prompt) — FIXED.** Root cause (reproduced: nezha bash -l -i→PROMPT_COUNT=2): .bash_profile carries the tmx-shell-init source line AND sources .bashrc which also carries it → one process double-sources; the blank/return path re-prompts. Fix: per-process non-exported _TMX_SHELL_INIT_PROMPTED guard in scripts/tmx-shell-init.sh.template. Test 54 RED→GREEN (3/3) + meta-test M-DBLPROMPT.
- **A42 (mouse copy) — FIXED.** Config was correct & live (not stale); copy pipe works. Cause: in mouse-tracking apps plain drag forwards to the app; iTerm2 Option Key Sends=Normal eats Alt-drag → only Shift-drag reached tmux (undiscoverable). Fix: prefix m mouse-toggle (native-terminal selection escape hatch) + Shift-drag docs in scripts/tmux.conf.template. Test 55 RED→GREEN (3/3) + meta-test M-MOUSEDTOGGLE; test 56 real-mouse honest SKIP (§11.4.3).
- **B3 (P5-M20/M21 escapes) — CLOSED + migrated to Fixed.md**
§B3. State-verified MUTATIONS CAUGHT 45 / ESCAPED 0 on Mistborn (tests 49/50 close the layer-4 gaps). M22 nezha CodeGraph baseline repaired via codegraph_setup.sh during dual-host setup.
- **Mistborn validation:** full suite PASS=51 FAIL=0 SKIP=4; meta-test 45 CAUGHT / 0 ESCAPED. Evidence in qa-results/ + plan at docs/superpowers/plans/2026-05-29-tmx-mouse-doubleprompt-b3-release.md.
- **Pending:** dual-host setup.sh (Mistborn redeploy + nezha pull+setup, nezha double-prompt re-prove, M22 repair),

CHANGELOG/changelog-doc with both-host evidence, VERSION bump (1.0.17/18), gh+glab release.

§3.1 Governance bring-up — Issues.md / Fixed.md / explicit §11.4.x anchors landed

Status: COMPLETE (2026-05-08T17:06Z).

- Created Issues.md (open / in-flight tracker) with:
 - Categories A-E, status conventions OPEN / PARTIAL / BLOCKED / RUNNING / INVESTIGATED
 - Status reclassification rules referencing §11.4.6
 - Seeded items: META-MUT-001, CHAL-COVER-001, TEST-AUDIT-001, TMX-T5, TMX-T7, TMX-T8, TOPO-DISPATCH-001
- Created Fixed.md (closed archive) with:
 - A0 initial migration (commit 08d4ba5, 2026-05-07)
 - B0 §1 covenant propagation (commit b92bf7f, 2026-05-08)
 - C0 test 09 crash-isolation-scope landed and green (commit b92bf7f, 2026-05-08)
- Updated Constitution.md:
 - Explicit §11.4.1, §11.4.2, §11.4.3, §11.4.4, §11.4.5 anchors added (preserving §1 prose verbatim)
 - §11.4.6 no-guessing mandate added with verbatim user-mandate quote
 - §9 absolute data safety added
 - §12.6 60% host memory budget added (TMX_MEM enforcement seam)
 - §5 expanded into full §12.10 continuation-document mandate
- Updated CLAUDE.md and AGENTS.md to carry every numbered anchor + canonical-authority cross-references

§3.2 Per-session containerization (Phase B — multi-session work)

Status: RESEARCH COMPLETE + WRAPPER LANDED + ISOLATION VERIFIED (2026-05-08).

Web research output: docs/plans/containerization.md recommends `systemd-run --user --scope (cgroup-v2 transient scope) over podman-per-session`. Wrapper at `scripts/tmx` implements `tmx {new|attach|ls|kill}` with `MemoryMax=$TMX_MEM` (default 8G), `CPUQuota=$TMX_CPU` (default 200%), `TasksMax=4096`, `Delegate=yes`.

Test 09 (scripts/tests/09_crash_isolation_scope.sh) — 14 PASS / 0 FAIL / 0 SKIP — verifies T1 (host capability), T2 (wrapper invariants), T3 (cgroup interface evidence), T4 (SIGKILL containment + user.slice survival), T6 (concurrent registration independence). Source-line breakdown:

- **T1:** systemd 258 + cgroup v2 mounted ✓
- **T2:** tmx wrapper invokes `systemd-run --user --scope + sets MemoryMax/CPUQuota/TasksMax/Delegate=yes` ✓
- **T3:** transient scope created; `/sys/fs/cgroup/.../memory.max` reads 268435456 bytes (matches set 256M) + `/sys/fs/cgroup/.../cpu.max` reads 50000 100000 (50% quota over 100ms period) — both POSITIVE EVIDENCE per §1 ✓
- **T4:** spawn scope, read `MainPID` from `cgroup.procs`, SIGKILL it, verify scope inactive AFTER kill, verify `default.target=active` THROUGHOUT (user.slice survives — Constitution §1 invariant) ✓
- **T6:** 3 concurrent scopes, all registered + active simultaneously ✓

§3.3 Pending tests — moved to Issues.md

The previously-listed deferred tests (T5 memory pressure under cap, TasksMax stress, Concurrent OOM independence) have migrated to Issues.md:

- **TMX-T5** — Issues.md C1 (PARTIAL; operator-unblock runbook for dedicated test host)
- **TMX-T7** — Issues.md C2 (OPEN; new test 11 planned)
- **TMX-T8** — Issues.md C3 (OPEN; new test 12 planned)

Cross-cutting blockers also tracked in Issues.md:

- **META-MUT-001** — Issues.md A1 (paired-mutation harness needed before §11.4.4 layer-4 coverage is real)
- **CHAL-COVER-001** — Issues.md B1 (HelixQA Challenge entries pending)
- **TOPO-DISPATCH-001** — Issues.md D1 (formal topology dispatch matrix)

§3.4 Anti-bluff enforcement — test audit + challenges fix + covenant propagation verification

Status: COMPLETE (2026-05-08T18:00Z).

- Verified anti-bluff covenant propagation across ALL governance files:
 - Root Constitution.md: §1 + §11.4.1-§11.4.6 present ✓
 - Root CLAUDE.md: full covenant present ✓
 - Root AGENTS.md: compact covenant present ✓
 - Containers/CLAUDE.md: full covenant present ✓
 - Containers/AGENTS.md: full covenant present ✓
 - tmux/ (upstream submodule pinned to 3.6a): N/A — no governance files, not modified per Constitution
- Performed line-by-line §11.4.2 anti-bluff audit of tests 01-09:
 - All 9 tests confirmed to carry positive runtime evidence (/proc files, cgroup interfaces, tmux command output, systemctl state).
 - No FAIL-bluff vectors found (all set -uo pipefail, guarded variables).
 - Full audit table moved to Fixed.md B2.
- Fixed broken paths in scripts/challenges/tmux.yaml (scripts/tmux/tests/ → scripts/tests/).
- Migrated TEST-AUDIT-001 from Issues.md B2 → Fixed.md B2.

§3.5 Hostname-derived status-bar colour — algorithm + wrapper + 2 tests + challenges

Status: COMPLETE (2026-05-08T19:00Z).

- Created scripts/hostname_color.sh — DJB2 hash over hostname → curated 27-colour palette. Deterministic, testable standalone.
- Updated scripts/tmx.template — added _apply_host_color() function that invokes hostname_color.sh and applies set -g status-style bg=<colour> on the running tmux server. Applied on both new-session and attach paths.
- Created scripts/tests/10_hostname_color_algorithm.sh — 5 invariants (deterministic, valid colourNNN, palette member, spread ≥12/16, empty-fallback to system hostname). All 5 PASS on this host.
- Created scripts/tests/11_hostname_color_integration.sh — verifies wrapper applies correct colour to tmux server. SKIPS if binary/wrapper not yet built.
- Updated scripts/tests/run_all.sh — glob changed from 0[1-9]_*.sh to [0-9][0-9]_*.sh to support tests 10+.

- Updated scripts/challenges/tmux.yaml — added TMUX-CH-10 (algorithm) and TMUX-CH-11 (integration).
- Updated docs/research/customization/colors.md — added implementation documentation covering algorithm, integration, verification, usage, and anti-bluff covenant.
- Updated AGENTS.md — bump test count from 9 to 11.

§3.6 Full coverage — missing challenge entries, destructive tests, topology dispatch

Status: COMPLETE (2026-05-08T20:00Z).

- **CH-09:** Added missing challenge entry for crash isolation scope (test 09) — was the only test without a challenge.
- **Test 12** (12_memory_pressure_under_cap.sh): TMX-T5 — allocates up to MemoryMax+10% inside a transient scope, captures dmesg OOM-kill evidence, validates user.slice survival. Gated by `TMX_TEST_DESTRUCTIVE=1`.
- **Test 13** (13_tasksmax_stress.sh): TMX-T7 — fork-bomb resistance. Spawns processes up to TasksMax=4096, reads pids.current/pids.max from cgroup interface. Gated by `TMX_TEST_DESTRUCTIVE=1`.
- **Test 14** (14_concurrent_oom_independence.sh): TMX-T8 — three scopes A/B/C, OOM-kills A, verifies B and C survive with original MainPIDs. Gated by `TMX_TEST_DESTRUCTIVE=1`.
- **CH-12/13/14:** Challenge entries for all three destructive tests.
- **Topology dispatch:** added `_probe_topology()` to scripts/tmx.template — detects systemd version + cgroup v2, classifies host as tmx-supported/tmx-degraded/tmx-unsupported per §11.4.3.
- **Fixed.md B2:** closure SHA updated to 68a65b0.
- **Issues.md → Fixed.md migration:** B1 (CHAL-COVER-001), C1 (TMX-T5), C2 (TMX-T7), C3 (TMX-T8), D1 (TOPO-DISPATCH-001) moved to Fixed.md.

§3.8 Audit cycle 2026-05-13 — submodule pin-drift caught + governance staleness fixed

Status: COMPLETE (2026-05-13T00:00Z).

Triggering event: user invoked `/init` followed by "what is left unfinished, no-bluff policy heavily enforced everywhere". Audit found and remediated:

- **CRITICAL §1 / §11.4.6 bluff:** f4132aa Auto-commit (2026-05-13) silently bumped tmux submodule from cc117b5 (tag 3.6a) to

3f651d9f (3.6a-329-g3f651d9f — 329 commits past the only stable tag). Governance docs across Constitution / CLAUDE / AGENTS / CONTINUATION / README still claimed "pinned to tag 3.6a". User decision: revert to tag (newer stable tag does not exist; 329 commits are unreleased upstream master). Submodule pointer reverted to cc117b5.

- **README.md staleness:** line 5 ("eight verification tests"), line 33 (PASS=6 FAIL=0 SKIP=2), line 37 ("8 tests cover"), line 39 (2-SKIP list) — all rewritten to current 14-test / PASS=10 SKIP=4 / 4-SKIP-list state with explicit `TMX_TEST_DESTRUCTIVE=1` callout and §11.4.4 layer-4 harness reference.
- **CLAUDE.md drift:** added top-of-file project summary + canonical-doc cross-links + fresh-conversation workflow; fixed `0N_*.sh` glob → `NN_*.sh`; named the four layers inline in the Test-interrupt rule; added "Files to never edit directly" section (parity with AGENTS.md).
- **AGENTS.md drift:** mirrored CLAUDE.md project summary + cross-links; fixed same `0N_*.sh` glob bug; added meta-test row to commands table.
- **Issues.md format:** restored A / B / C / D / E section headers as empty sections (was: A / B / E only, with C and D missing despite conventions list referencing them). All bodies note "landed in Fixed.md" with item IDs for traceability.
- **CONTINUATION.md timestamp:** 2026-05-08T22:00Z → 2026-05-13T00:00Z; this entry added under §3.8 per §12.10 invariant.
- **M6 mutation added** (`scripts/tests/meta_test_false_positive_proof.sh`): injects `$$` into the `hostname_color` hash, making the algorithm non-deterministic per-invocation. Test 10 T1 must FAIL under this mutation. Closes the audit-flagged gap: until M6, the "same-host = same-color" user invariant was protected by side-effect only (no dedicated anti-randomness mutation).
- **docs/guide/README.md phantom-script bluff fixed:** 3 occurrences each of `verify_tmux.sh` → `verify.sh`, `setup_tmux.sh` → `setup.sh`, `install_tmux_deps.sh` → `install_deps.sh`; "8 tests" → "14 tests" (2×).
- **GUIDE.md "severity hierarchy" pre-existing bluff caught** (Fixed.md A3): documented "blockers / critical / advisory" classification does not exist in `run_all.sh` — every test is treated equally (any FAIL = RED). Rewritten to describe the actual gate logic. Caught while extending the table for tests 09-14; would have been silently doubled otherwise.
- **Verification + validation cycle 2026-05-13** (Fixed.md A14): operator invoked `superpowers:verification-before-completion` +

asked for full anti-bluff verification + governance propagation. Every verification command run FRESH this session per Iron Law. Captured evidence: `setup.sh --verify-only SUMMARY=PASS=10 FAIL=0 SKIP=5 GREEN`; `test_e2e.sh PASS=9/0/0 GREEN`; test 15 `PASS=6/0/0`; live `tmx new -s VerifyDemo` shows operator's host shell (`milosvasic@Mistborn`, which `brew=/opt/homebrew/bin/brew`, `ulimit -t=86400`, `ulimit -u=2666`). Two defects caught and fixed: (D1) wrapper hostname resolution drifted to FQDN after bridge removal — restored `scutil-LocalHostName` path so colour stays `colour202` for `Mistborn`; (D2) `install_deps.sh` + `build_oom_set.sh` weren't OS-aware out of the box — now `install_deps.sh` detects Darwin → `brew install` (no `sudo`), `build_oom_set.sh` SKIPS on non-Linux. Governance propagation completed: verbatim user-mandate quote now in root `CLAUDE.md` + `AGENTS.md` (previously only in `Constitution` + `Containers`); §11.4.7 reference added to `Containers/CLAUDE.md` + `Containers/AGENTS.md` (previously only `Containers/CONSTITUTION.md`). Install state: shell snippets in both `~/.bashrc` + `~/.zshrc`, `zsh -c 'which tmx; tmx -V'` returns native wrapper + `tmux 3.6a`. macOS `RLIMIT_AS` gap documented honestly per §1 (XNU `EINVAL` for unprivileged `setrlimit` on memory).

- **Per-session isolation + Constitution §11.4.7** (Fixed.md A13): user reported `core@localhost` + lack of OOM containment. Forensic: `tmx new -s isol1 -d + tmx new -s isol2 -d` previously shared ONE `cgroup` (`run-p504653-i504654.scope`) — README's "if one session OOMs, others survive" was a §1 bluff. Architectural rewrite: each `tmx new -s NAME` now spawns its OWN `tmux` server on socket `tmx-NAME` inside its OWN scope `tmx-NAME.scope` with host-adaptive `MemoryMax` (`max(MemTotal × 60% / 4, 2GB)`), `CPUQuota=200%`, `TasksMax=4096`, `Delegate=yes`. OOM in any session = contained to that scope; B+C survive with original `MainPIDs` (verified test 14 `PASS=8/0/0` with `stress-ng tmx send-keys`). Test 15 NEW: per-session `cgroup` distinctness (6 assertions, positive readback from `/sys/fs/cgroup`). Test 11 + test 14 + test 08 rewritten to OPERATOR PATH (`tmx new -s NAME` instead of hand-crafted `systemd-run --user --scope`). e2e T7 NEW: bridge confirms two distinct scopes via VM `systemctl`. Meta-test M5 retargeted, M7 retargeted, M9+M10 NEW (10/10 mutations caught). Plan + operator decisions captured in [docs/plans/per-session-isolation.md](#). **Constitution §11.4.7 NEW** (operator-path test coverage rule): every gate test MUST exercise the same entry point an end-user invokes; tests that hand-craft equivalents are supplementary. Propagated to `Containers/CONSTITUTION.md` at same depth. Forensic anchor: Fixed.md A12 (status-bar green default) + A13 (sessions sharing one `cgroup`) — both shipped while non-operator-path tests reported GREEN.

- **Regression protection for A10 — gate-coverage gap closed**

(Fixed.md A11): user demanded "make sure nothing passes again! zero-bluff policy MUST BE followed blindly!" — honest accounting of why the existing gates missed the colour bug:

- test 11 always passed -S "\$SOCKET" → tested explicit-socket path only; operator's default-socket use case uncovered.
- meta-test M4/M5 targeted scripts/tmx (the SSH bridge on Darwin) instead of scripts/tmx-vm (the actual VM wrapper) — paired-mutation harness was mutating the wrong file.
- test_e2e.sh pre-fix didn't read status-style at all.
- Triple-layer regression protection added: (1) test 11 T6 — new test exercising default-socket path with positive evidence (tmux show -g status-style with NO -S, FAILs explicitly on bg=green default-applied bluff); (2) meta-test M4/M5 retargeted to scripts/tmx-vm; (3) M7 (re-introduce SOCK-empty early return) + M8 (hardcode bg=green) both caught by test 11 T4.1/T6. Result: all 8 mutations caught + reverted (16 PASS), full destructive suite PASS=14/0/0, e2e PASS=8/0/0.

- **Status-bar colour silently green (default) + bridge ignored**

macOS hostname (Fixed.md A10): user said "Bottom was green for current host. Is that expected for mistborn?" — two stacked bluffs caught:

- _apply_host_color / _apply_oom_score had [-n "\$sock"] || return 0 — silently bailed when tmx new -s Test was invoked WITHOUT -S (which is the primary operator use case; tmux uses its default socket then). So set -g status-style was never fired → tmux default bg=green shown. Test 11 always passes -S "\$SOCKET" so the bug was uncovered. Fix: helpers now build target=(-S "\$sock") only when sock is non-empty; otherwise omit -S and let tmux use its default socket.
- Bridge scripts/tmx-mac.template didn't forward macOS hostname → hostname_color.sh inside VM resolved against localhost.localdomain regardless of macOS host. Every macOS user got the same VM-derived colour, violating host-distinguishability. Fix: bridge captures scutil --get LocalHostName (e.g., "Mistborn"), forwards as TMX_HOSTNAME=<host> in the remote SSH command. Wrapper's _apply_host_color passes \${TMX_HOSTNAME:+"\$TMX_HOSTNAME"} to hostname_color.sh — env var is source of truth when set, \$ (hostname) fallback otherwise.
- New test_e2e.sh T4.5 explicitly reads tmx show -g status-style, computes expected colour from scutil --get LocalHostName (or hostname on Linux), FAILs on bg=green (default-applied bluff) or non-match. Captured runtime

evidence on Mistborn: status-bg 'colour202' matches
hostname-derived 'colour202' for 'Mistborn'.

• **Interactive tmx new fixed + e2e automation harness** (Fixed.md A9):

- User ran `tmx new -s` Test interactively after install and hit open terminal failed: not a terminal. The bridge + `ssh -t` allocated TTY correctly; the WRAPPER's `cmd & wait` pattern was disconnecting tmux from the foreground process group → no TTY access. Detached-mode tests (`-d`) all passed because tmux daemonizes; interactive (no `-d`) hit the bluff. §11.4.1 FAIL-bluff in the wrapper itself.
- Fix: wrapper now detects `-d/-D` in args (INTERACTIVE flag), and for interactive: runs `systemd-run` in foreground inheriting TTY, injects `-d` after the new-session keyword (so tmux server detaches), then `exec attach` to take over TTY interactively. For detached: keeps existing path.
- New: `scripts/test_e2e.sh` — 7-test operator-facing automation: prerequisites + `tmx -V` (via bridge) + new-session + send-keys + capture-pane + survive-without-client + kill-session. All with positive runtime evidence; trap-on-EXIT cleanup. Captured the literal marker string echoed inside the session pane through the bridge — proves the full operator stack carries intent end-to-end.
- Verified: `bash scripts/test_e2e.sh GREEN 7/0/0`;
`TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh` still GREEN 14/0/0 (wrapper change didn't regress existing tests).

• **macOS tmx fully operational + side-by-side end-to-end** (Fixed.md A8):

- `bash scripts/setup.sh` on Darwin now diverges intelligently: builds (containerized), generates BOTH `scripts/tmx-vm` (Linux wrapper for VM paths) AND `scripts/tmx` (Darwin SSH bridge from new `scripts/tmx-mac.template`); runs `scripts/test_vm.sh` for §11.4 target-env verification (since binary cannot exec on Darwin); on GREEN, appends `bashrc` snippet to BOTH `~/.bashrc` AND `~/.zshrc` (macOS defaults to `zsh`).
- **Bridge mechanics:** `scripts/tmx-mac.template` discovers podman machine's SSH endpoint via `podman machine inspect` at every call (port re-discovery — survives `podman machine stop && start`), verifies `tmx-vm` is executable in the VM, then `ssh -t -i <identity> -p <port> core@127.0.0.1 "<vm-repo>/scripts/tmx-vm <quoted-args>"`.
- **Verified end-to-end on Darwin 24.5.0 / Apple Silicon:** `zsh -c 'source ~/.zshrc; which tmx; tmx -V' → /Users/.../scripts/tmx + tmux 3.6a`; `system /opt/homebrew/bin/tmux -V → tmux 3.6a`

— side-by-side coexistence proven; VM-side verify GREEN
PASS=11 FAIL=0 SKIP=3.

- **scripts/verify.sh now respects \$WRAPPER and \$TMUX_BIN env vars** so test_vm.sh / bridge tooling can redirect to the right wrapper without source modifications.
- **README.md** gained a new "Architecture" section with ASCII diagram showing Linux-native vs Darwin-bridge dispatch; docs/guide/README.md §5.5 documents the bridge mechanics + Mermaid flowchart; CLAUDE/AGENTS one-liners updated.
- **Final sweep — env-specific wrapper + \$255 violations + sed portability** (Fixed.md A7):
 - **Environment-bound wrapper:** scripts/tmx was generated for one env (container /repo paths OR VM /Users/.../ paths) and silently failed in the other. New scripts/test_vm.sh orchestrator regenerates the wrapper with VM-native paths before each VM-side run. Both test_containerized.sh and test_vm.sh now own the wrapper-regeneration step for their respective target.
 - **Constitution \$255 violations:** "ATMOSphere" in GUIDE.md title + body, challenge yaml description, oom_set.c license comment, tmux.conf.template attribution; atm_tmux_test_* / atm_test_* / atm_tmx_test_color_* socket/session names across 8 test files; ~/.tmux.conf.pre-atmosphere backup naming. All renamed: code uses tmx_* / pre-vasic-digital; docs branded "vasic-digital".
 - **sed -i portability:** setup.sh used GNU-only sed -i for bashrc markers — would fail on macOS BSD sed. Centralized in _strip_bashrc_snippet() using perl -i -ne / ... flip-flop range delete (portable GNU+BSD).
 - **Verified post-fix in VM:** TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh → PASS=14 FAIL=0 SKIP=0; META=1 bash scripts/test_vm.sh → 12/12 mutations caught.
- **Install-mechanism side-by-side bluffs** (Fixed.md A6): user asked about the alias and side-by-side coexistence requirement. Audit found three real defects:
 - bashrc snippet pointed PATH at scripts/tmux/ (phantom subdir; wrapper is at scripts/tmx).
 - alias tmux='tmx' line shadowed the system tmux command → broke side-by-side.
 - setup.sh closing message said "Then 'tmux' will use the ATMOSphere build" (wrong command name + stale branding).
 - **Canonical answer to user's question:** the alias is **tmx** (not tmux). Wrapper at scripts/tmx generated from tmx.template. tmx new|attach|ls|kill invokes the verified vasic-digital tmux;

tmux invokes whatever was on the operator's PATH before (system / Homebrew / apt / dnf). Both coexist.

- **Full destructive + meta-test cycle caught 6 more FAIL-bluffs** (Fixed.md A5): pushing past PASS=10 SKIP=4 into actually running TMX_TEST_DESTRUCTIVE=1 + meta-test in the podman machine VM (Fedora CoreOS 42 + systemd 257 + stress-ng + tmx-oom-set setcap) surfaced multiple §11.4.1 FAIL-bluffs. Final: **PASS=14 FAIL=0 SKIP=0** (full verify) + **12/12 mutations caught** (meta-test). Real defects fixed:

- Tests 12/14 _skip;<continue> bug — printed SKIP but didn't exit; continued running and FAILED on missing stress-ng → false §11.4.1 FAIL-bluff. Fix: explicit exit 0.
- Tests 12/14 unprivileged dmesg → fallback to journalctl -k via _kring_count / _kring_tail helpers; broader OOM regex.
- Test 13 fork-storm OOM-killed its own scope (4096 sleeps × 700KB > 256M MemoryMax); two-phase scope, polling registration, test TASKS_MAX 4096→256 with MemoryMax=512M (production wrapper still TasksMax=4096 — separately grep-verified by test 09 T2.2).
- Meta-test sed -i strip-exec-bit → orig_mode capture/restore via stat + chmod.
- Meta-test M5 sed pattern was eval-expanded (\${TMX_MEM: -8G} → 8G) so it never matched → silent no-op → mutation escaped (a **bluff in the gate itself**). Fix: simplified to literal MemoryMax=|MemMax= pattern.
- Meta-test had no debug visibility → added opt-in TMX_META_DEBUG=1.
- **The user's "same-host = same-color" invariant is now PROVEN end-to-end** with tmux show -g status-style reading colour166 on first session AND on second session on same host (test 11 T4.1 + T5).

- **Build pipeline bluffs caught while reproducing on macOS** (Fixed.md A4): three real defects surfaced when user asked why we couldn't build on macOS:

- docker/Dockerfile: groupadd -g 20 collided with Ubuntu's dialout group → build aborted at step 7/10. Fix: -o (non-unique) on groupadd + useradd. Build now reproducible on Darwin hosts.
- docker/build_inside_container.sh: LDFLAGS -ljemalloc was being dropped by linker's default --as-needed because tmux doesn't reference jemalloc symbols. **README's "Build-time -ljemalloc" claim was a §1 bluff until this commit**. Fix: -Wl,--no-as-needed -ljemalloc -Wl,--as-needed. Verified ldd now shows libjemalloc.so.2 in DT_NEEDED.

- `docker/build_inside_container.sh:make -j2` silently no-op'd after LDFLAGS change (existing binary defeated mtime check).
Fix: prepend `make clean` so LDFLAGS changes always take effect.

The f4132aa Auto-commit itself is a §12.10 / §11.4.6 violation — opaque commit message, silent submodule mutation, no CONTINUATION update. Cannot rewrite history per §9 data safety; documenting here is the audit trail.

§3.7 §11.4.4 layer-4 paired-mutation harness landed (A1 META-MUT-001)

Status: COMPLETE (2026-05-08T21:00Z).

- Created `scripts/tests/meta_test_false_positive_proof.sh` — §11.4.4 layer-4 harness with 5 registered mutations:
 - **M1:** break `hostname_color` output format → test 10 T2 FAILs (invalid format)
 - **M2:** force hash to zero → test 10 T3 FAILs (zero spread)
 - **M3:** single-entry palette → test 10 T3 FAILs (palette mismatch)
 - **M4:** remove `systemd-run` flag from `tmx.template` → test 09 T2 FAILs
 - **M5:** remove `Delegate=yes` from `tmx.template` → test 09 T2 FAILs
- Results on this host: **10 PASS / 0 FAIL / 0 SKIP** — all mutations caught.
- Updated `CLAUDE.md`: removed "PENDING" references, wired mandatory operations to the actual file.
- Updated `AGENTS.md`: layer 4 now marked as landed, not PENDING.
- Updated `Fixed.md`: A1 entry with closure details; all "PENDING META-MUT-001" references replaced with actual coverage references.
- Updated `Issues.md`: A1 removed (→ `Fixed.md` A1). `Issues.md` now has **zero open items** — all originally seeded items are closed.
- Cleaned up `CONTINUATION.md` §3.6: removed stale "remaining open" note.

§3.16 — v1.0.15 multi-line + paste-IN + alt-screen Claude Code + workable-items SQLite + DOCX + constitution sync (2026-05-28)

Status: IN PROGRESS — Mistborn build GREEN + Mistborn full suite GREEN + meta-test GREEN; nezha deploy pending; release tag pending.

Triggering event: operator mandate (verbatim, 2026-05-28):

"In last iteration of work we did we have allegedly fixed copy / paste from and to the tmux session window. Selecting multiple lines and copying of them does not work. We MUST BE able to scroll vertically everywhere and copy / paste anything! Especially in Claude Code (claude command)! Do in depth investigation and systematic debugging, reproduce issue and then apply comprehensive fixes! ... Every bug / issue we report MUST BE first covered with all supported tests which will actually confirm failure / problems we report! Then, after we apply fixes, the same tests MUST CONFIRM all changes and fixes really working! We MUST for every issue or change, for any workable item create proper workable item in our Issues doc and all relevant docs around it (with them all being exported in all expected file types - PDF, HTML, DOCX)! We MUST NEVER forget the flow: workable item → SQLite database → all docs we have related to workable items!"

Five-PWU pipeline executed per §11.4.58 + §11.4.94 (parallel-by- default + zero-idle). Phase 0 fetched + ff-merged constitution/ from 84c948d → 6828ff2 (19 commits) and Containers/ from fbef9d6 → 2e9ca0e (17 commits). PWU-A (tmux conf + 4 tests + 4-layer coverage) + PWU-B (§11.4.87..98 propagation) + PWU-C (Go binary FULL Phase 3+ project-local) + PWU-D (DOCX export) ran in parallel; PWU-E (commit + deploy) is the closing sequential PWU.

- **A37 — Multi-line + PASTE-INTO + alt-screen TUI mouse-drag override** (Fixed.md A37): the user's primary ask. Identified the three uncovered surfaces vs v1.0.14: (1) multi-line drag in Claude-Code-like TUIs (mouse_any_flag=1 forwards drag to app), (2) paste-INTO from OS clipboard (set-clipboard external is copy-OUT only), (3) the alt-screen + mouse-tracking surface never asserted end-to-end. Added M-/S-MouseDrag1Pane overrides + matching drag- end @clip-routed binds + @clip-read user-option + prefix+P paste bind. Tests 45/46/47/48 land 4-layer coverage with pbpaste-back physical evidence: PASS=6/0/0 + 6/0/0 + 8/0/0 + 9/0/0 = 29 PASS, RED→GREEN proven for tests 46 + 48. Synthetic alt-screen surrogate synthetic_alt_screen_app.py substitutes for Claude Code in tests (no OAuth flake per §11.4.98). Mutations M46 + M48 CAUGHT + RESTORED.
- **A38 — Constitution + Containers sync + §11.4.87..98 propagation** (Fixed.md A38): submodule pointers advanced via --ff-only; short-form mirrors for 12 new universal anchors landed in project Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md (each

anchor appearing ≥ 3 times in each consumer per PWU-B audit).

New Layer-1 gate L1C in verify.sh enforces literal presence.

- **A39 — Workable-items SQLite SSoT (Go binary, project-local Phase 3+)** (Fixed.md A39): constitution scaffold ships Phase-2 stubs only; project added cmd/workable-items/ with full Phase 3+ logic (sync md-to-db / sync db-to-md / diff / validate / add / close / report). 11 Go sources + embedded schema (drift-check-cited from constitution 6828ff2) + 10 tests passing 30/30 (3 iters per §11.4.50). Pure-Go modernc.org/sqlite (no CGO). Initial DB at docs/workable_items.db seeded with 45 items (TMX-001..TMX-045) from live Issues.md + Fixed.md. TRACKED in git per §11.4.95. Honest gaps logged: (a) legacy items default to Type=Task (no Type lines pre-§11.4.16); (b) live-corpus round-trip not byte- identical for free-form bodies (per §11.4.93 phase 6); (c) upstream PR + integration into commit_all.sh deferred.
- **A40 — DOCX export** (Fixed.md A40): scripts/sync_all_markdown_ exports.sh extended to emit .docx siblings alongside .html + .pdf via pandoc. 44/44 candidates produced valid DOCX. Layer-1 gate CM-DOCX-EXPORT-SYNC enforces canonical-doc sibling presence.

Multi-host deploy + release plan (PWU-E, sequential):

1. ✓ Mistborn setup.sh --rebuild GREEN.
2. ✓ Mistborn full suite via verify.sh — **PASS=45 FAIL=0 SKIP=3** (3 honest topology SKIPS: oom_score_adj Linux-only, destructive-gated, nezha-required).
3. ✓ Mistborn meta-test — 43 CAUGHT / 2 ESCAPED (pre-existing P5-M20+P5-M21 from v1.0.9) / 8 SKIPPED.
4. ✓ Mistborn go test ./cmd/workable-items/... -count=3 — 30 PASS / 0 FAIL.
5. commit_all.sh push (this commit).
6. Nezha SSH (~/nezha script) → cd ~/Projects/tmux && git fetch --all && git pull && bash scripts/setup.sh --rebuild && bash scripts/setup.sh --verify-only — full gate capture expected. T5 honestly SKIPS on headless Linux per §11.4.3 — T2/T3/T4 still PROVE the binding chain.
7. Release tag v1.0.15 via GitHub + GitLab CLIs with comprehensive docs/changelogs/v1.0.15.md referenced.
8. Post-release version bump (per operator mandate "After releasing... increase version (version code) and commit and push all work").

§3.15 — v1.0.14 clipboard copy-OUT physical proof + e2e stale-prereq fix + multi-host deploy (2026-05-22)

Status: COMPLETE (2026-05-22).

Triggering event: operator mandate (verbatim, 2026-05-22):

"We can always copy / paste from and to the terminal window and current tmux (tmx) session! Using mouse or keyboard MUST WORK properly!!! Scrolling the content / history MUST NOT be broken or anything else! After all changes are done, covered with full validation and verification tests make sure we setup new version to current host (macOS) and nezha (nezha.local is back online). Once both hosts are updated with the latest versions and on both hosts all tests are executed with success with no failure and bringing real proofs of everything working as requested and expected (physical proofs) with no bluffs commit and push all Submodules and main repo to all upstreams and release new version with comprehensive in-depth version (change) log using GitHub and GitLab CLIs."

- **A35 clipboard copy-OUT physical proof** (Fixed.md A35): identified the §101 PASS-bluff hole — bindings existed and grepped fine, but no test ever read back the OS clipboard.
 - NEW scripts/tests/44_clipboard_copy_out_physical.sh — operator-path. Spawns tmx new -s NAME -d, prints a unique marker, enters copy-mode, search-backward + select-line, invokes @clip (T3 direct -X + T4 the literal y keystroke that triggers the bind table end-to-end), then reads the OS-native paste tool (T5 — pbpaste / wl-paste -n / xclip -o / termux-clipboard-get) and asserts the marker is there. T5 SKIPS honestly with reason on headless Linux; T3/T4 binding- chain proof runs everywhere so the test is never inert. Pre-test save + post-test restore of the operator's clipboard.
 - PASS=7/0/0 this cycle (Darwin); T5 returned the marker via pbpaste — physical end-user evidence.
 - Layer-1 verify.sh static gate extended (4 new _l1 checks for @clip + the three bindings).
 - Layer-3: TMUX-CH-44 in scripts/challenges/tmux.yaml.
 - Layer-4: M44 in meta_test_false_positive_proof.sh strips the @clip definition line — test 44 T1 catches universally (structural grep), T5 additionally catches where a clipboard tool is reachable. CAUGHT + RESTORED both directions.
- **A36 e2e stale podman-machine prerequisite** (Fixed.md A36): scripts/test_e2e.sh T1.2 still hard-required a running podman

machine, a leftover from the pre-v1.0.7 SSH-bridge architecture. Fixed by probing for the native Mach-O binary FIRST and only falling back to the podman check for bridge-era installs. e2e GREEN after the fix.

- **B3 P5 escape transparency** (Issues.md B3): meta-test reported 2 ESCAPES on Mistborn — P5-M20 (non-TTY guard) and P5-M21 (cwd-capture hook). Nezha additionally reports M22 as an ENVIRONMENTAL escape (CodeGraph state). All three pre-exist this cycle. Investigation: P5-M20/M21 are layer-4 test-DESIGN gaps — the assertions cannot distinguish "guard fired" from "fallback fired", so a single-layer strip slips through. M22 is an environmental state-dependency on the CodeGraph index/config on nezha. Underlying features GREEN. Surfaced transparently in Issues.md B3 + the v1.0.14 CHANGELOG — anti-bluff disclosure per §11.4 / §101 / §11.4.6. Closure conditions documented; deferred to a future cycle.
- **Multi-host deploy** (this turn's operator ask, satisfied): Mistborn updated to v1.0.14 via `setup.sh --rebuild`. Nezha fetched gitlab/main, fast-forwarded to v1.0.14, rebuilt and installed. Both hosts run the same code; both gates GREEN; both e2e GREEN; clipboard physically proven on Mistborn (pbpaste) and binding-chain proven on nezha (tmux buffer, T5 honest-SKIP on headless).

§3.14 — v1.0.8 uniform tmux UI recolouring (active border + clock + window-status-current) (2026-05-21)

Status: COMPLETE (2026-05-21T21:00Z).

Operator-driven follow-up to v1.0.7: "flying animated top decoration"

- clarification "Do coloring of all UI tmux parts with proper color we use instead of default green. Anything colored with that green colors has to become the color we have assigned to the bottom view we are coloring."

Pre-v1.0.8 `_apply_host_color()` set only `status-style bg=$color`. Other default-green tmux surfaces (active pane border, clock face, selected-window highlight) stayed green. v1.0.8 extends the wrapper to apply the hostname-derived colour atomically to all four:

- `status-style` `bg=$color`
- `pane-active-border-style` `fg=$color`
- `clock-mode-colour` `$color`
- `window-status-current-style` `bg=$color,fg=black`

mode-style + message-style deliberately untouched (default yellow, not green; yellow provides best contrast against any palette bg).

NEW test 26 (26_ui_color_uniformity.sh) — operator-path session spawn + live show -gv readback per surface — PASS=5/0/0 on Darwin.
NEW M24 paired mutation — regex-strips the three v1.0.8 set-lines from the generated wrapper, asserts test 26 T2/T3/T4 FAILs. MUTATION CAUGHT + FEATURE INTACT both directions.

Captured operator-path readback (Mistborn, this session):

- status-style: bg=colour44
- pane-active-border-style: fg=colour44
- clock-mode-colour: colour44
- window-status-current-style: bg=colour44, fg=black → All four surfaces uniformly turquoise (colour44 = RGB 0,215,215).

Verification (Darwin quintuple-fresh, 2026-05-21):

- setup.sh --verify-only → PASS=24 FAIL=0 SKIP=2
- meta-test → 36 caught / 0 escaped / 6 skipped
- e2e → PASS=9 FAIL=0 SKIP=0
- codegraph_validate → PASS=4 FAIL=0 SKIP=1

Released v1.0.8; setup.sh re-run as final install per operator "Make sure we setup the new version once all done." directive.

§3.13 — v1.0.7 cross-host portability + hostname-colour rebalance + Node-22 pin + release (2026-05-21)

Status: COMPLETE (2026-05-21T18:30Z).

Operator-driven cycle: tried to install on Nezha (Linux), 3 tests FAILED → 6 fix rounds → Nezha GREEN → operator reported orange-color collision (Nezha + Mistborn) → palette rebalanced + test 25 + M23 mutation → release v1.0.7.

Six fix rounds (all Nezha-driven; live forensic capture):

1. Initial 3-fix attempt (test 09 timing, test 17 ingestion race, test 21 codegraph bootstrap). Test 17 fixed; tests 09 + 21 had secondary defects.
2. Bash -lt fractional bug in round-1 poll-loop (silent fail); npm-prefix PATH augmentation for non-interactive shells.
3. CodeGraph init clobbers config.json on every version; auto-update wiring per §11.4.80 mandate ("ALWAYS latest").

4. cgroup.procs sweep (kill all PIDs not just first) + `stat -f '%z'` portability bug (Darwin vs Linux).
5. Test 09 T4.2 rewritten to verify the REAL containment invariant (cgroup.procs drained), not systemd's unit-state transition (which is sticky on systemd 258 / ALT 11 — verified live).
6. macOS Node 22 LTS pin: codegraph 0.8.0 refuses Node 25 per upstream issue #81. Operator authorised `brew install node@22 && brew link --force --overwrite`. Updated .zshrc references.

Then the hostname-colour palette rebalance:

- Operator: "nezha and Mistborn both show orange". Palette had 7 orange-family colours out of 27.
- Rebalanced to 27 entries spanning the hue spectrum; no two adjacent within RGB Euclidean distance 80.
- NEW test 25: T1 nezha+Mistborn distance ≥ 80 (post-fix: 332.7); T2 16-synthetic-hostname pairwise within pigeonhole tolerance; T3 palette adjacency check.
- NEW M23 mutation: revert palette → test 25 FAILs.

Verification (Darwin quintuple-fresh, 2026-05-21):

- `setup.sh --verify-only` → PASS=23 FAIL=0 SKIP=2
- `meta-test` → 34 caught / 0 escaped / 6 skipped
- `e2e` → PASS=9 FAIL=0 SKIP=0
- `codegraph_validate` → PASS=4 FAIL=0 SKIP=1
- `codegraph_cadence_check` → GREEN

Nezha verification: round-5 `setup.sh` full pipeline GREEN (after fix rounds 1-5 landed). v1.0.7 includes all 6 rounds + palette + test 25.

Out-of-scope this cycle (logged honestly per §11.4.6):

- Auto-detection of Node version compatibility in `setup.sh` — deferred to v1.0.8.
- Pre-tag Nezha re-verify of v1.0.7 final → will run after tag push as deferred post-release smoke.

§3.12 — v1.0.6 workable-items closure + §11.4.80 cadence + Containers/QWEN.md + release tag (2026-05-21)

Status: COMPLETE (2026-05-21T15:30Z).

Operator directive (verbatim): "Finish all workable items you have presented here, rebuild and install again, retest fully and if fully working as expected release new version!"

Closed (this cycle):

- **§11.4.80 automatic-trigger wiring** (was the one deferred item from v1.0.5 audit). New `scripts/codegraph_cadence_check.sh` + `scripts/codegraph_install_cadence.sh`. Darwin `launchd` plist installed + loaded (verified via `launchctl list`). Cross-platform git pre-push hook installed (warns on STALE; `CADENCE_MODE=block` available). Anti-bluff: cadence check reads STAMP CONTENT (regenerated_at + node_count), not just existence. §11.4.81 cross-platform-parity dispatch in the installer: Linux gets `systemd` user timer; Darwin gets `launchd` plist.
- **Containers/QWEN.md covenant gap** (was the second-to-last out-of-scope item from v1.0.5 — `Containers/QWEN.md create`). Containers commit `fbef9d6` pushed to github + gitlab. Parent pointer bumped `4ca5491` → `fbef9d6` in this cycle's commit.
- **Full rebuild + install** via `bash scripts/setup.sh` (not just `--verify-only`). `~/tmux.conf` installed; `.bashrc` + `.zshrc` snippets appended; `tmx` wrapper installed.
- **Quintuple-fresh verification** captured this session: `setup.sh --verify-only` → PASS=22/0/SKIP=2; `meta-test` → 32 caught / 0 escaped / 6 skipped; `e2e` → 9/0/0; `codegraph_validate` → 4/0/SKIP=1; `codegraph_cadence_check` → GREEN.

Still out of scope (logged honestly per §11.4.6):

- Cadence-script paired §1.1 mutation (backdate-stamp → cadence FAILs) — deferred to v1.0.7. The script's content-based read is already anti-bluff; the mutation would tighten regression protection.
- CodeGraph upstream `--include-submodules` — separate cycle per §11.4.74.
- Linux-host CI runner — would exercise the Linux `systemd` user timer install path + the Linux branches of tests 09/13/14.

Release v1.0.6 tagged + pushed per §11.4.40 (complete fresh retest preceded the tag; no spot-check shortcut).

§3.11 — §11.4.81 cross-platform-parity + Darwin branches + constitution §11.4.81 (2026-05-21)

Status: COMPLETE (2026-05-21T07:30Z).

Triggering events (in arrival order):

- Operator (2026-05-21): "Any Linux-only blocker / issue we have MUST BE created macOS and other supported platforms equivalent! So, depending on platform proper implementation will be used for particular OS! EVERYTHING MUST BE PROPERLY EXTENDED AND UPDATED!"
- Operator (2026-05-21): "Add this OS / Platform details / rules / mandatory constraints into our root (constitution Submodule) Constitution.md, CLAUDE.md, AGENTS.md, QWEN.md and other constitution Submodule relevant files so all projects who inherit these in the future immediately in such situations create proper missing implementations for particular platform / OS! Make sure you first fetch and pull the latest version of constitution Submodule. Once all done and extended properly commit and push constitution Submodule to all upstreams, then re-process and re-evaluate all rules and mandatory constraints..."
- (Power blackout mid-cycle — work resumed cleanly from working tree.)

Plan + execution captured at docs/plans/v1.0.5.md. Seven phases all landed in one v1.0.5 commit per §11.4.42 iteration discipline.

- **§11.4.26 step 1+:** constitution submodule fetched + ff-merged from 7f738df → 19ce1b1 (brought in §11.4.79 + §11.4.80 CodeGraph anchors).
- **§11.4.81 anchor LANDED in constitution submodule (pushed 6e164f3 to origin HelixDevelopment).** Universal cross-platform-parity mandate + mirror blocks in Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md. Three sub-mandates: (A) per-OS implementation REQUIRED via runtime dispatch; (B) per-OS tests REQUIRED with positive captured evidence per branch; (C) honest kernel-gap citation + adjacent equivalent test REQUIRED where no equivalent exists. constitution/QWEN.md ALSO gained the verbatim 2026-04-28 anti-bluff covenant quote (audit gap fix).
- **A26 — §11.4.79 compliance fix.** Removed constitution/** and Containers/** from .codegraph/config.json exclude (own-org MUST be INCLUDED); kept tmux/** excluded (third-party). scripts/

codegraph_validate.sh NEW — 5 probes (V1 CLI version, V2 node count, V3 §11.4.79 split, V4 honest-gap re submodule traversal, V5 MCP spawn). M22 paired mutation (re-exclude → V3 FAILs). Honest gap: CodeGraph 0.6.8 doesn't traverse submodules; config compliance met but practical cross-submodule indexing waits for upstream CodeGraph support.

- **A25 + A22 + A24 — Darwin branches for tests 09/13/14 + NEW test 24 (per §11.4.81 just landed).** All four exercise the macOS POSIX rlimit primitives that are the kernel-enforced equivalents of the Linux cgroup primitives:
 - **Test 09 Darwin (PASS=6/0/0):** wrapper invokes `tmx-rlimit-wrapper.sh`, two operator-path sessions, distinct server PIDs, `ulimit -t/-u` readback inside each pane via `send-keys + capture-pane`, `SIGKILL` session A's server, session B survives with ORIGINAL PID (positive evidence per §11.4.5).
 - **Test 13 Darwin (PASS=2/0/0):** child bash lowers `ulimit -u 64`
 - fork-bombs; captures `EAGAIN` occurrences from `stderr` (bash: fork: Resource temporarily unavailable) = XNU kernel-enforced `RLIMIT_NPROC`.
 - **Test 14 Darwin (PASS=5/0/0):** 3 operator-path sessions; `SIGKILL` session A's server (macOS adjacent test for OOM-independence per §11.4.81 (C) — Darwin has no OOM killer); verify B+C survive with ORIGINAL PIDs + `tmx ls` still lists them.
 - **NEW Test 24 Darwin (PASS=2/0/0):** child bash sets `ulimit -t 2`
 - CPU-bound loop; verifies process killed by signal 24 (`SIGXCPU`) after ~3s wall (exit rc=152 = 128+24); also verifies `TMX_CPU_HARD_SEC=7200` propagates to `RLIMIT_CPU=7200` inside session. §11.4.81 (C) adjacent test for the XNU `RLIMIT_AS` gap.
- **A25 paired mutations refresh.** M7-M10 RETIRED (targeted dead `scripts/tmx-vm` VM-wrapper path, replaced by native dual-OS per Fixed.md A4-A8). M20 + M21 NEW for Darwin `rlimit`. M22 for §11.4.79 `codegraph` exclude. Meta-test on Darwin: 32 caught / 0 escaped / 6 skipped (M4/M5 topology-correct SKIP on Darwin; M7-M10 retired-with-rationale SKIPS).
- **Constitution submodule pointer bumped** 19ce1b1 → 6e164f3 in this same project commit per §11.4.26 step 7.

Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --verify-only PASS=22/0/SKIP=2, meta-test 32/0/SKIP=6` (all SKIPS §11.4.3 topology-correct or retired-with-rationale), e2e 9/0/0, `codegraph_validate PASS=4/0/SKIP=1`.

Out-of-scope this cycle (honest tracking per §11.4.6):

- §11.4.80 automatic-trigger wiring (cron / git hook for the constitution-provided `codegraph_update.sh` + `codegraph_sync.sh`) — deferred. Manual invocation works today.
- CodeGraph upstream `--include-submodules` support — out of scope per §11.4.74.
- Containers/QWEN.md create — separate PR to vasic-digital/Containers per §11.4.28.
- Linux-host CI runner — would let us exercise the Linux branches of tests 09/13/14 alongside the Darwin branches running today.

§3.10 — CodeGraph (§11.4.78) + verbatim covenant + AUDIT fixes + docs reorg (2026-05-21)

Status: COMPLETE (2026-05-21T05:30Z).

Triggering events (in arrival order):

- Operator (2026-05-21): create full working plan for tackling all these points completely; all existing tests + Challenges MUST work anti-bluff; covenant MUST be part of project + submodule Constitution / CLAUDE / AGENTS; HelixConstitution submodule is the root.
- Operator (2026-05-21): incorporate / install CodeGraph for Claude Code, OpenCode, Kimi CLI, Crush, Qwen Code; comprehensive anti-bluff tests.
- Operator (2026-05-21): full documentation + user guides + HTML + PDF exports.
- Operator (2026-05-21): do NOT modify constitution submodule, only keep it regularly updated.
- Operator (2026-05-21): documentation under context-named subdirectories per the constitution rule.

Plan + execution captured at `docs/plans/v1.0.4.md`. Eight phases all landed in one v1.0.4 commit per §11.4.42 iteration discipline.

- **A18 CodeGraph integration (§11.4.78).** CLI v0.6.8 installed; `.codegraph/config.json` tracked with §11.4.10 secret exclusions + §11.4.28 owned-submodule paths; `.codegraph/codegraph.db`

gitignored; §11.4.77 regen manifest + scripts/
codegraph_reindex.sh. MCP wired for all 5 agents (Claude Code
.mcp.json NEW; OpenCode + Kimi audited; Crush .crush.json NEW;
Qwen Code .qwen/settings.json NEW). All configs reference bare
codegraph on PATH. Comprehensive docs/codegraph/README.md.
Tests 20/21/22, challenges CH-20/21/22, mutations M16/M17.

- **A19 verbatim covenant propagation.** §11.4 2026-04-28 user
mandate now LITERALLY present in project CLAUDE.md + AGENTS.md +
QWEN.md (was only in Constitution.md). Verify.sh Layer-1 gate added;
test 19 PASS=7/0/0; CH-19; M15 (mutates a temp copy — real
CLAUDE.md is never touched).
- **A20 AUDIT-1 fix.** M4/M5 meta-test mutations were silently
SKIPping on Darwin for the wrong reason (BSD sed -i quirk). Real
root cause: the mutations target Linux-only cgroup/systemd-run
wrapper code unreachable on Darwin. Fix: explicit uname -s
topology guard per §11.4.3 + portable inplace_sed for Linux runs.
- **A21 AUDIT-2 fix.** tmx kill -t NAME (documented friendly verb) was
ambiguous because tmux only knows kill-pane/kill-server/ kill-
session/kill-window. Wrapper now translates the bare kill verb to
kill-session. Test 23 PASS=5/0/0; CH-23; M19.
- **Docs reorganised** under context-named subdirectories per the
operator's invocation of the constitution rule: 7 docs moved
(GUIDE/SCROLLING/CODEGRAPH/CONTAINERIZATION_PLAN/
NATIVE_DUAL_OS_PLAN/ PER_SESSION_ISOLATION_PLAN/
PLAN_v1.0.4). Every reference in governance + scripts updated
atomically.
- **§11.4.65 universal-Markdown export.** New scripts/
export_docs.sh (pandoc HTML + weasyprint PDF, idempotent, per-
file timeout 60s) refreshed siblings for every consumer Markdown.
- **§11.4.71 pre-push.** Parent + constitution/ + Containers/ all at
upstream tip; no divergent commits.
- **Honest gaps logged (§11.4.6) — out of scope this cycle:**
 - Upstream constitution/QWEN.md covenant insert → separate PR
to HelixDevelopment/HelixConstitution (operator forbids
modifying constitution from inside this project).
 - Containers/QWEN.md create + populate → separate PR to vasic-
digital/Containers per §11.4.28.

- Shell parser for CodeGraph (currently only the C file indexed, 6 nodes) → upstream contribution to add tree-sitter shell.
- Agent-driven unforgeable-challenge end-to-end test → classified AUTONOMOUS_DEIGNED per §11.4.52 carve-out (mechanical seam exists via test 22 T7; agent-driven layer lands when a headless agent harness is wired).

Full gate this cycle (Darwin arm64, 2026-05-21): `setup.sh --verify-only PASS=18/0/SKIP=5, meta-test 26/0/SKIP=6` (all SKIPS §11.4.3 topology-correct), e2e 9/0/0.

§3.9 — Scrolling fix + HelixConstitution inheritance (2026-05-21)

Status: COMPLETE (2026-05-21).

Triggering event: operator research note — tmux scrolling (up/down of terminal output, especially inside the Claude Code TUI) did not work; requirement to scroll vertically from ANY computer or mobile phone (Termux/Android). Plus the mandate to incorporate the HelixConstitution submodule as the root of the Constitution / CLAUDE.md / AGENTS.md, inherited further.

- **A16 scrolling fix** (Fixed.md A16): `scripts/tmux.conf.template` — `history-limit 2000→50000`, `mode-keys vi`, `WheelUp/WheelDown` overridden to always drive copy-mode scrollbar (works on desktop mouse + trackpad + Termux touch-scroll), `allow-passthrough + extended-keys + terminal-features extkeys` for the Claude Code TUI, OS-adaptive `@clip clipboard`. NEW test 17 (operator-path, PASS=13/0/0), `verify.sh` Layer-1 static gate, TMUX-CH-17, M12 + M13 paired mutations.
- **A17 HelixConstitution** (Fixed.md A17): added the `constitution/` submodule (pinned 7f738df). Full refactor of `Constitution.md` to the extends-template form (Project Articles §101–§109); `CLAUDE.md` / `AGENTS.md` inheritance pointers; new `QWEN.md`. NEW test 18 (PASS=10/0/0), TMUX-CH-18, M14 + CM-CONSTITUTION-INHERITANCE paired mutations. The constitution mutation runs on a TEMP COPY — the real, decoupled `constitution/` submodule is never modified (operator directive).
- **Containers submodule**: adopted remote 4ca5491, which already carries HelixConstitution recursive inheritance (`find_constitution.sh`, `QWEN.md`, all four governance docs). Parent gitlink bumped b077f2c → 4ca5491.

- **Meta-test portability:** M1/M2/M3/M6 converted from GNU sed -i to a portable inplace_sed helper — they now run on macOS instead of silently skipping. Meta-test: 18 caught / 0 escaped / 6 skipped (was 10/0/10).
- Full gate this cycle (Darwin arm64, 2026-05-21): setup.sh --rebuild GREEN, suite PASS=13/0/SKIP=5, meta-test 18/0/6, e2e 9/0/0.

§4 — Recent commits

- (this commit, 2026-05-21) — **A16 scrolling fix + A17 HelixConstitution inheritance (v1.0.3 / versionCode 4).**
tmux.conf.template: history-limit 50000, vi copy-mode, WheelUp/Down → copy-mode scrollbar override, Claude Code passthrough, OS-adaptive clipboard. HelixConstitution submodule added at constitution/; Constitution.md refactored to the extends-template form (§101–§109); CLAUDE/AGENTS inheritance pointers + new QWEN.md. Tests 17 + 18, challenges CH-17/18, mutations M12–M14 + CM-CONSTITUTION-INHERITANCE; meta-test M1/M2/M3/M6 made portable. Containers gitlink → 4ca5491. Gate: PASS=13/0/5, meta 18/0/6, e2e 9/0/0. See Fixed.md A16 + A17, §3.9.
- (this commit, 2026-05-16) — **A15 cosmetic .exe-strip cycle (v1.0.2 / versionCode 3).** Bottom-left status bar showed claude.exe because Claude Code v2.x ships its macOS native binary literally as .../anthropic-ai/claude-code/bin/claude.exe (verified Mach-O 64-bit ARM64). Patched scripts/tmux.conf.template to set automatic-rename-format "#{s/\\.exe\$//:pane_current_command}" — literal-dot-anchored, escape-verified-empirically. Added test 16 (operator-path per §11.4.7, includes regression guard binary t16_bashe to catch the unescaped-dot bug class), challenge TMUX-CH-16, M11 paired mutation. Live operator-path validation recorded: pane_current_command='claude.exe' → #W='claude'. Full gate: PASS=11 FAIL=0 SKIP=5 (same SKIP profile as v1.0.0). See Fixed.md A15.
- (commit before that, 2026-05-13) — Audit cycle: tmux submodule reverted from 3f651d9f to cc117b5 (3.6a tag) after silent pin-drift caught; README.md test-count + PASS/SKIP staleness fixed (8→14 tests, PASS=6→10, SKIP=2→4); CLAUDE.md + AGENTS.md gained project summary, cross-links, fresh-conversation workflow, NN_*.sh glob fix, named four layers, "Files to never edit directly"; Issues.md restored A/B/C/D/E section headers; CONTINUATION.md §3.8 + timestamp refresh per §12.10. See §3.8.
- f4132aa — Auto-commit: opaque submodule bumps (Containers e377dea→af51968 legitimate upstream governance; tmux

cc117b5→3f651d9f reverted in this commit). §12.10 / §11.4.6 violation documented in §3.8.

- (previous commit) — §11.4.4 layer-4 paired-mutation harness (META-MUT-001): meta_test_false_positive_proof.sh with 5 mutations (10/0/0 all caught); CLAUDE.md/AGENTS.md PENDING→LANDED; Issues.md now empty of open items. Also: fixed 5 broken script references (REPO_ROOT wrong depth, filenames); implemented systemd-run --user --scope wrapping in tmx.template; test 09 topology dispatch fixed (functional probe, not mount grep); CLAUDE.md compacted to match AGENTS.md; binary built and verified GREEN (PASS=10 FAIL=0 SKIP=4: OOM helper + 3 destructive).
- 39284ab — Full coverage: CH-09 added (was missing); tests 12/13/14 (T5/T7/T8 destructive) with TMX_TEST_DESTRUCTIVE=1 gate; topology probe in tmx.template; challenge entries CH-12/13/14; B1/C1/C2/C3/D1 migrated Issues→Fixed.
- 8b37046 — Hostname-derived status-bar colour: DJB2→27-colour palette algorithm + wrapper integration + tests 10/11 + challenges + docs.
- 68a65b0 — Anti-bluff enforcement: TEST-AUDIT-001 complete (9 tests §11.4.2-audited), challenges paths fixed, AGENTS.md compact rewrite, covenant propagation verified.
- b92bf7f (2026-05-08) — §1 anti-bluff covenant verbatim user-mandate quote added; test 09 crash-isolation-scope (14/0/0) landed.
- 08d4ba5 (2026-05-07) — Initial vasic-digital/tmux: migrated from ATMOSphere project + per-session containerization plan + 8-test verification gate.

§8 — Resumption recipe

1. cd ~/Projects/tmux
2. Read this document
3. Read Constitution.md, CLAUDE.md, AGENTS.md, Issues.md, Fixed.md
4. Find the topmost IN PROGRESS / OPEN item, resume
5. Update this document in the SAME commit as the work itself (§12.10 invariant)